K-Bound: When Is Label-Free Adaptation Knowable?

Helpful, Harmful, and Unknowable Regimes Under Distribution Shift

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Abstract

An unlabeled adaptation procedure can produce an update without establishing that deploying it will help. We study when the available evidence supports ADAPT, FREEZE, or ABSTAIN for a fixed model pair. For binary 0/1 loss and positive disagreement mass, K-Bound characterizes the exact compatible-benefit interval under a declared bound β on a hidden target calibration residual. In the full label-kernel class, a strict direction is uniformly justified exactly when $|M| > \beta$, where M is the observable disagreement margin. A complementary evidence-fibre theorem gives the smallest residual bound that any uniformly valid audit of the same information must cover. Additional processing cannot remove ambiguity left by that information. Knowability-Guided Adaptation (KGA) addresses a separate empirical target: the measured score difference between a candidate and its frozen baseline on an evaluation cell. Labeled development and disjoint residual-calibration cells fit its benefit predictor and error interval; the action uses no scored-cell outcome. On the opened, dependent six-family CIFAR-10-C diagnostic, Tent and EATA have lower point regret than both fixed policies. Tent makes 2 false ADAPT decisions among 1,091 ADAPT actions and 1 false FREEZE among 337 FREEZE actions; EATA makes neither error among 1,207 ADAPT and 105 FREEZE actions. SAR instead favors always-adapt. On CCT-20, all 45 cells retain the frozen model, including one false FREEZE; five-checkpoint DomainNet development evaluation yields universal abstention. Thus these natural-shift studies support conservative retention, not useful selective routing. The theory distinguishes structural ambiguity from prediction uncertainty; the experiments do not establish coverage or population protection on unseen shifts. **Keywords:** test-time adaptation; adaptation gating; abstention; partial identification; distribution shift; conformal calibration; selective decisions.

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1 Introduction

Test-time adaptation changes a model using unlabeled deployment data. Entropy minimization, filtering, regularization, and related methods can recover accuracy under shift. The same update can also make predictions worse under unfavorable shifts (Wang et al., 2021; Niu et al., 2022, 2023; Wang et al., 2022). The central deployment question is therefore directional action correctness: an adapter supplies a candidate, not a reason to deploy it.

Our primary research question is: *When can evidence available without new target labels justify replacing a frozen predictor with a fixed adapted candidate?* Let f_0 be the frozen predictor and f_a the candidate. Their population benefit is

$$\Delta = R_T(f_0) - R_T(f_a). \tag{1}$$

A positive value favors the update; a negative value favors the frozen model. We use three decisions. ADAPT asserts positive benefit, FREEZE asserts negative benefit, and ABSTAIN asserts neither. Both non-adapt decisions serve f_0, but only FREEZE makes a directional claim.

K-Bound treats this as partial identification. In a binary model under 0/1 loss, we fix the input law, the predictors, and every label-free observable. We then vary the target label kernel within a declared class. The compatible benefits give an exact frontier for uniform strict decisions. The class is indexed by a bound on a hidden target calibration residual. That bound is an assumption, not a quantity supplied by an unlabeled batch. A second theorem shows that an audit using the same evidence must cover the residual range that the evidence leaves unresolved.

Knowability-Guided Adaptation (KGA) is the empirical companion. It predicts a model pair's measured benefit on an evaluation cell. It acts only when a residual-calibrated interval excludes zero. KGA is label-free at the scored decision, not during development and calibration: labeled development cells train the predictor, and disjoint residual-calibration cells calibrate its errors. The specified cross-fit excludes each scored cell's outcome from its own prediction, radius, and decision. This computational separation does not make an already opened retrospective panel prospective; outcome access is reported separately for each study. K-Bound is the population partial-identification framework; KGA is the empirical cell-benefit interval gate. They share three-action decision logic, but not a target or guarantee: K-Bound reasons about population benefit Δ through (M, γ, β), whereas KGA predicts measured cell benefit Δ^{cell} through $(\widehat{\Delta}, \varepsilon)$.

1.1 Three Scientific Contributions

1. **Exact population frontier.** We construct target label laws with identical label-free evidence, derive their feasible benefit interval, and characterize its strict positive, strict negative, and unresolved regions, including the zero-benefit boundary.
2. **Evidence-fibre audit limit.** We identify the least residual bound that the same observable law can support uniformly, explaining why historical labels help only through a stated link to the deployment domain.
3. **Empirical paired-benefit gate.** We give a cell-outcome-disjoint three-way cross-fit for the controlled grids and evaluate its ADAPT/FREEZE/ABSTAIN behavior on constructed shifts and its retention behavior on natural shifts. We report action exposure and false-action denominators before secondary accuracy and regret consequences. The standard coverage-to-action implication states exactly what follows if the interval covers its declared target.

The opened CIFAR-10-C diagnostic first asks whether each committed direction was correct and reports the corresponding action exposure. Tent makes two false ADAPT actions and one false FREEZE; EATA makes neither; SAR makes no FREEZE decisions. The secondary utility result is candidate-specific: Tent and EATA have lower point regret than either fixed policy, whereas SAR favors always-adapt. CCT-20 instead preserves frozen-model utility in all 45 cells with no ADAPT exposure. This is conservative frozen-model retention, not selective routing on an unseen natural shift. The appendices retain adverse, incomplete, and withheld studies so that this boundary does not depend on removing unfavorable evidence.

Newer painting-development evaluations retain the same limitation: the five-checkpoint cohort and separate reference-source candidates yielded universal abstention, with adverse queue and normalization-state diagnostics preserved as development evidence.

Sections 3–5 define the population target, derive the strict-commitment frontier, and establish the evidence-fibre audit floor. Section 6 introduces the separate empirical gate. Sections 7–8 report the controlled CIFAR diagnostic and CCT-20 retention result. Section 9 states the assumptions, missing evidence, and deployment limits.

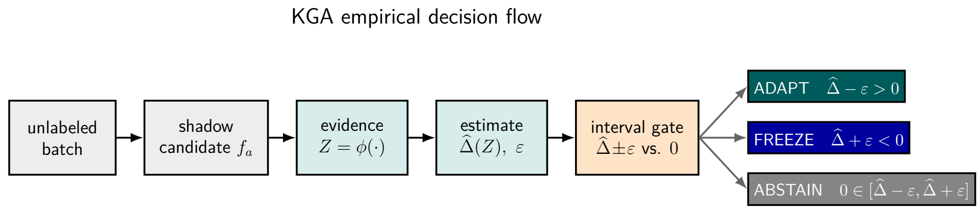

Figure 1: KGA's empirical decision flow for one evaluation cell. A shadow adapter creates f_a from frozen f_0. Labeled development cells fit the benefit predictor, and disjoint residual-calibration cells determine its error radius. The scored cell contributes label-free evidence only. An interval above zero selects ADAPT, one below zero selects FREEZE, and an interval crossing zero selects ABSTAIN. The figure defines the action rule; it does not establish interval coverage.

2 Related Work

2.1 Candidate Adaptation and Update Selection

Tent, EATA, SAR, CoTTA, MEMO, SHOT, and test-time training construct candidates through entropy objectives, filtering, regularization, temporal stabilization, augmentation, or auxiliary tasks (Wang et al., 2021; Niu et al., 2022, 2023; Wang et al., 2022; Zhang et al., 2022; Liang et al., 2020; Sun et al., 2020). Other methods decide where or when to update. TTSA selects modality-specific adapters (Chen et al., 2025). GALA selects layers and filters samples by gradient alignment (Sahoo et al., 2025). Informed Adaptation learns topological features associated with useful updates (Mutlu et al., 2026). KGA starts later in this pipeline: it compares one realized candidate with its frozen baseline. Learned routing and rollback are established ideas, not our novelty claim.

2.2 Label-Free Performance and Strategy Estimation

ATC, agreement-on-the-line, AETTA, and prediction-matrix statistics estimate performance from unlabeled outputs and suitable development information (Garg et al., 2022; Baek et al., 2022; Miller et al., 2021; Lee et al., 2024; Deng et al., 2023; Xie et al., 2024). The closest empirical comparator is Kim et al. (2024), whose TTA/ALine method uses agreement-based estimators after adaptation, supports hyperparameter selection, and compares adaptation strategies without labeled OOD data.

KGA directly fits the paired score difference for one frozen/candidate pair and converts its prediction interval into a three-way action. Separate adapted- and frozen-model accuracy estimates can also be differenced; a synchronized current-policy comparison with official TTA/ALine and AETTA implementations remains missing.

2.3 Monitoring and Adaptation Controllers

POEM protects adaptation through entropy matching and online betting (Bar et al., 2024). Schirmer et al. (2025) monitor an adapting model's running risk against a source-risk reference under a proxy-informativeness condition; a supervised precursor uses a labeled test stream (Podkopaev and Ramdas, 2022). Drift-to-Action Controllers use delayed labels and sequential evidence to gate adaptation, abstention, rollback, and retraining (Lamaakal et al., 2026). These methods act during an evolving stream. KGA instead decides whether one proposed candidate should be committed before it replaces the frozen state; it makes no anytime-valid risk claim.

2.4 Partial Identification and Robust Decisions

An identified set contains the values compatible with the observations and assumptions. Decision theory under partial identification studies actions when that set leaves payoffs ambiguous. Christensen et al. (2026), for example, combine worst-case payoff uncertainty with statistical learning of identified parameters. K-Bound asks the narrower question of whether one strict benefit sign is uniform over the set. It neither solves the general optimal-decision problem nor claims interval thresholding as a new idea.

Indistinguishable problems are already central to domain-adaptation impossibility (Ben-David et al., 2010a,b). Our contribution is a disagreement-conditional label-kernel construction, the resulting clipped benefit set, and its residual-audit floor. The proof does not introduce label-free impossibility. Risk-estimation assumptions based on conditional independence (Steinhardt and Liang, 2016), label shift (Lipton et al., 2018; Garg et al., 2020), or disagreement discrepancy (Rosenfeld and Garg, 2023) restrict the compatible laws. Likewise, MMD-credal risk intervals impose RKHS and covariate-shift structure (Ariq, 2026). Each restriction can shrink the identified set. A converse theorem for a richer class does not contradict those results.

2.5 Marginal and Conditional Coverage

Selective classification and reject-option learning abstain on individual predictions (Geifman and El-Yaniv, 2017; Chow, 1970; Cortes et al., 2016). KGA instead abstains on an update. Its coverage-to-action proposition uses standard interval logic (Tibshirani et al., 2019; Barber et al., 2023; Bates et al., 2021; Angelopoulos et al., 2025). The statistical burden is to justify coverage for the intended unit, especially after selection or under shift.

Jin and Ren (2025)'s JOMI procedure targets coverage conditional on selection, assuming exchangeability and a compatible selection rule. Gibbs et al. (2025) obtain conditional guarantees over specified shift classes through a different calibration construction. KGA's ordinary residual interval implements neither method. Its marginal false-adapt implication must therefore remain separate from error conditional on ADAPT and from coverage within each target environment. Learn Then Test addresses the separate problem of configuration selection and multiplicity (Angelopoulos et al., 2025).

3 Setup, Notation, and Claim Levels

This section fixes the probability target, evaluation unit, model pair, and gate before defining actions or error events. Table 2 separates the population and empirical symbols used throughout.

3.1 Model Pair, Evidence, and Evaluation Unit

Let $(\Omega, \mathcal{F}, P_T)$ support a target draw (X, Y), and let P_S denote the source law. For loss ℓ, define

$$R_T(f) = \mathbb{E}_{P_T}[\ell(f(X), Y)].$$

We fix a realized pair of measurable predictors for the population comparison. The frozen predictor is f_0; an adapter $\mathcal{A}$ uses an unlabeled batch $X_{1:m}$ to construct the fixed candidate $f_a = \mathcal{A}(f_0, X_{1:m})$. Population risk then integrates a fresh target draw while holding this pair fixed.

The gate may read label-free evidence

$$Z = \phi(X_{1:m}, f_0, f_a),$$

such as predictions, logits, and update statistics. It may not read the outcome of the cell it is about to score.

Scope of the matched-world observation experiment. The population evidence-law and randomized-decision statements use the same fixed observation experiment in each compatible world and the same fixed decision kernel. The observation law factors through the fixed input marginal; finite iid input sampling is one proved realization. Other dependent sampling designs require their own common-law premise. Randomized decisions require a common joint law of observation and random seed; an independent common seed is one sufficient construction, whereas equal seed marginals alone do not establish that joint law. Equality of one-input

Table 1: Neighboring approaches differ in inferential target, decision timing, and guarantee. The comparison is conceptual; the archived empirical ports are not an official synchronized benchmark.

Approach	Target	Decision timing	Guarantee relevant here
TTA/ALine; AETTA	adapted-model performance	after candidate evaluation	performance estimate
POEM; risk monitors	running adaptation risk	during or after adaptation	sequential monitoring under named assumptions
Partial-identification decisions	payoff set across compatible laws	decision under ambiguity	robust-decision context
K-Bound / KGA	population benefit set / predicted cell benefit	before candidate commitment	strict frontier / coverage-conditional action

Table 2: Population and empirical quantities. β bounds a population residual by assumption; ε is an empirical prediction-error radius. They are not interchangeable.

Quantity	Level	Meaning
f_0, f_a	both	frozen predictor and realized shadow candidate
Δ	population	expected risk reduction $R_T(f_0) - R_T(f_a)$
$D,\ \mu_T(D)$	population	disagreement region and its target probability
M	population	label-free mean correctness-score margin on D
$\gamma,\ \beta$	population	hidden calibration residual and its externally declared absolute bound
Δ^{cell}	empirical	observed score difference on one predeclared evaluation unit
$\widehat{\Delta},\ \varepsilon$	empirical	predicted cell benefit and residual-based interval radius
FA_u	empirical	probability of ADAPT and a nonpositive cell benefit
FA_c	empirical	probability of nonpositive cell benefit conditional on ADAPT
FF_u	empirical	probability of FREEZE and a nonnegative cell benefit
FF_c	empirical	probability of nonnegative cell benefit conditional on FREEZE
Commitment rate	empirical	fraction of ADAPT or FREEZE decisions; not interval coverage

marginals does not imply equality for arbitrary world-dependent batch couplings. These are the formal model's premises, not conclusions about the reported data. Holding the model pair fixed in this experiment does not justify conditioning on a pair adapted on that same batch. We do not establish that the transductive benchmark satisfies this observation model: outcome blindness alone supplies neither that conditioning argument nor iid evaluation.

An *evaluation cell* j is a predeclared collection of records $\mathcal{E}_j$ on which both models are scored. For a fixed higher-is-better score S, its measured benefit is

$$\Delta_j^{\text{cell}} = S(f_a; \mathcal{E}_j) - S(f_0; \mathcal{E}_j).$$

This is not the population benefit Δ. For accuracy it is an average of paired correctness differences. For macro-F1 it is a difference between two cell-level scores, not an average of per-example losses. We therefore report metric scales separately.

3.2 Actions and Their Operational Meaning

Definition 1 (Decision semantics)**.** The gate g maps its permitted evidence and frozen calibration artifacts to $\{\text{ADAPT}, \text{FREEZE}, \text{ABSTAIN}\}$. ADAPT commits f_a when an interval supports positive benefit. FREEZE rejects f_a when an interval supports negative benefit. ABSTAIN retains f_0 when neither strict direction is supported or the assessment is unavailable.

Freeze is a supported rejection; abstention is an unresolved assessment. An operator may attach monitoring, label acquisition, or review to an abstention. The experiments here evaluate only the frozen fallback and uncertainty record, not those additional interventions. If evidence is missing, retaining f_0 must be logged as ABSTAIN when the interface returns an unresolved assessment. An integrity failure may instead raise an exception; the caller must preserve the frozen service and record the failure, not a certified FREEZE. Algorithm 1 describes the decision semantics after this integrity boundary has been handled.

The observed-cell oracle chooses the model with the larger score. Regret is the policy's score shortfall from that oracle, averaged over the named cells with the stated weights. Always-freeze regret is $\max(\Delta_j^{\text{cell}}, 0)$; always-adapt regret is $\max(-\Delta_j^{\text{cell}}, 0)$. KGA serves the candidate only after ADAPT. Lower regret is better. A zero-benefit tie has zero regret for either model but supports neither strict sign.

3.3 Commitment, Interval Coverage, and Error

Let J be a random evaluation unit drawn from a predeclared deployment-unit distribution Q. The common probability space includes fitting and calibration randomness, candidate construction, J, and the records in $\mathcal{E}_J$. Write $\Delta^{\text{cell}} = \Delta_J^{\text{cell}}$ and $g = g_J$. The probabilities below are prospective marginal quantities under that joint law. When we instead summarize a finite recorded grid, we use the equal-weight empirical measure over its cells and call the result descriptive. It does not estimate the prospective law without an additional sampling design.

With this law and the gate fixed, we distinguish three measurements:

$$\text{Commit} = \mathbb{P}(g \in \{\textsc{adapt}, \textsc{freeze}\}), \quad (2)$$

$$\text{Cov}_{\text{cell}} = \mathbb{P}(|\widehat{\Delta} - \Delta^{\text{cell}}| \le \varepsilon), \quad (3)$$

$$\text{FA}_{\text{u}} = \mathbb{P}(g = \textsc{adapt}, \Delta^{\text{cell}} \le 0). \quad (4)$$

Conditional false adaptation is $\text{FA}_{\text{c}} = \mathbb{P}(\Delta^{\text{cell}} \le 0 \mid g = \textsc{adapt})$, defined only when $\mathbb{P}(g = \textsc{adapt}) > 0$. The operational proposition below bounds only the marginal error, and only when the stated interval coverage premise holds. It does not bound FA_{c}. A gate that never adapts has zero false adaptations mechanically, so every error report includes action exposure. A false FREEZE is defined analogously by $g = \textsc{freeze}$ and $\Delta^{\text{cell}} \ge 0$. Its conditional rate is undefined when no FREEZE action occurs.

The population theorem uses binary labels and 0/1 loss. The empirical gate can target a predeclared scalar score difference, including a multiclass metric, but that interface does not extend the binary theorem. The reviewer guide in Appendix L records the multiclass identity.

Six-symbol guide. M is the observable population-side disagreement margin; γ is the hidden target calibration residual; and β is an externally declared bound on $|\gamma|$. These quantities bracket population benefit Δ. At the empirical level, Δ^{cell} is the measured cell benefit, $\widehat{\Delta}$ is its prediction, and ε is the calibrated prediction-error radius. Thus (M, γ, β) supports population reasoning, whereas $(\widehat{\Delta}, \varepsilon)$ brackets Δ^{cell}; the two uncertainty objects are not interchangeable.

4 Population Partial-Identification Frontier

The decision depends on one unknown sign: the sign of Δ. Target labels hide that sign. M supplies the visible direction, γ records what labels still hide, and β limits how large that hidden term may be. Together they yield the exact three-action frontier.

4.1 Disagreement-Region Model

The two predictors can differ in risk only where they disagree. We therefore isolate that region before introducing the score and its hidden residual.

Assumption 1 (Deployment setup)**.** Let $Y \in \{0, 1\}$ and use 0/1 loss. Fix measurable binary predictors f_0 and f_a. Their disagreement region is $D = \{x : f_0(x) \ne f_a(x)\}$; assume $\mu_T(D) > 0$.

Let $s : \mathcal{X} \to [0, 1]$ be a fixed measurable candidate-correctness score fitted from source information. The target law has the measurable correctness kernel $\eta_a(x) = \mathbb{P}_T(f_a(X) = Y \mid X = x)$. Conditional on D, define

$$\begin{aligned} M &= \mathbb{E}[s(X) \mid D] - \tfrac{1}{2}, \\ \gamma &= \mathbb{E}[\eta_a(X) - s(X) \mid D]. \end{aligned} \quad (5)$$

Table 3: Validity obligations for the paper's three claim levels. Each row states its target, required premise, supported conclusion, and excluded inference.

Claim level	Required premise	Supported conclusion	Not implied
Population frontier	binary 0/1 loss, fixed augmented evidence, declared residual class; richness for necessity	exact strict-sign frontier within that class	learning β from unlabeled data
Operational proposition	marginal interval coverage for a named scalar target and unit	marginal false commitment for that target	conditional, group, or anytime control
Empirical evaluation	fixed candidates, splits, metric, and action rule	measured cell regret, inclusion, and action exposure	population transfer or prospective confirmation

The class radius $\beta \geq 0$ is supplied externally. It defines $\mathcal{C}_\beta = \{P_T : |\gamma(P_T)| \leq \beta\}$. Because $s \in [0, 1]$, the feasible margin is $M \in [-\frac{1}{2}, \frac{1}{2}]$. The identities below do not assume that s is source-calibrated.

Remark 1 (Residual identity). By definition, $M + \gamma = \mathbb{E}[\eta_a(X) \mid D] - \frac{1}{2}$. This is an algebraic decomposition, not evidence that calibration transfers. We call γ the target disagreement-conditional calibration residual; conditioning can make it nonzero even without distribution shift. The reviewer guide in Appendix L gives the full counterexample and calibration qualification.

Definition 2 (Risk alignment)**.** Evidence is risk-aligned when evidence-identical laws cannot have opposite nonzero benefit signs. This is weaker than uniform strict-sign identification: a fibre may still contain zero and positive benefit. We use the augmented evidence $\widetilde{Z} = (Z, M)$.

Definition 3 (Strict directional soundness)**.** At a fixed augmented-evidence law, ADAPT is sound when $\Delta(P) > 0$ for every compatible $P \in \mathcal{C}_\beta$, and FREEZE is sound when $\Delta(P) < 0$ throughout the fibre. A pointwise maximal rule commits exactly when one strict action is sound and abstains otherwise; $\Delta = 0$ supports neither strict claim.

4.2 Label-Kernel Freedom on the Disagreement Region

The central obstacle is now visible. Labels can change on D while the input distribution and all model outputs stay fixed. The next lemma isolates that freedom.

Lemma 1 (Label-kernel freedom on disagreement)**.** *Fix $(\mu_T, P_S, f_0, f_a, s)$, and therefore fix D and M. Also fix the conditional label law outside D. For every measurable $\eta : D \to [0, 1]$, there is a target law with input marginal μ_T and $\mathbb{P}(f_a(X) = Y \mid X = x) = \eta(x)$ on D. Across these laws, every label-free observable formed from the input batch, fixed models, and model outputs under the declared common observation experiment has the same distribution. Meanwhile,*

$$\gamma = \mathbb{E}[\eta(X) \mid D] - \tfrac{1}{2} - M.$$

Proof. On D, one binary predictor outputs 0 and the other outputs 1. Assign conditional probability $\eta(x)$ to $Y = f_a(x)$ and $1 - \eta(x)$ to $Y = f_0(x)$. Retain the fixed label kernel outside D. This construction changes neither the input marginal nor any model output. The declared observation law factors through that fixed marginal, so its label-free evidence laws are preserved. Equation (5) gives the displayed expression for γ. □

Consequently, every statistic formed from target inputs and fixed model outputs under this observation experiment has unchanged law, although the models' relative correctness can change.

4.3 Benefit Reduction and Matched-Evidence Impossibility

The first result connects the decision target to the three population symbols. Because the models agree off D and binary correctness is complementary on D, benefit depends only on $M + \gamma$.

Lemma 2 (Disagreement-region reduction). *Under Assumption 1, with* $\bar{a} := \mathbb{P}_T(f_a{=}Y \mid D)$*,*

$$\Delta = 2\,\mu_T(D)(\bar{a} - \tfrac{1}{2}),$$
$$\operatorname{sign}\Delta = \operatorname{sign}(\bar{a} - \tfrac{1}{2}) = \operatorname{sign}(M + \gamma).$$

The identity $\operatorname{sign}\Delta = \operatorname{sign}(M+\gamma)$ *holds for* every *target law; it does not assume* $|\gamma| \le \beta$*.*

Proof. Outside D, the two predictors agree and incur the same 0/1 loss. On D, exactly one is correct. The adapted and frozen accuracies conditional on D are therefore $\bar{a}$ and $1-\bar{a}$. Hence $\Delta = 2\mu_T(D)(\bar{a} - \frac{1}{2})$. The definitions of M and γ give

$$\begin{aligned} M + \gamma &= \Big(\mathbb{E}_{\mu_T|D}[s] - \tfrac{1}{2}\Big) + \mathbb{E}_{\mu_T|D}[\eta_a - s] \\ &= \mathbb{E}_{\mu_T|D}[\eta_a] - \tfrac{1}{2} = \bar{a} - \tfrac{1}{2}. \end{aligned}$$

Because $\mu_T(D) > 0$, the three quantities have the same sign. ☐

The reduction tells us which hidden quantity controls the sign. The harder question is identification. Can two target worlds agree on every permitted observation while assigning opposite signs to Δ? Inside the declared band, the answer is yes.

Theorem 1 (Interior matched-evidence impossibility). *Fix* $\beta > 0$*. For any feasible margin* $M \in [-\frac{1}{2}, \frac{1}{2}]$ *with* $|M| < \beta$*, there exist* $P_T^+, P_T^- \in \mathcal{C}_\beta$ *with the following properties:*

$$\operatorname{Law}(\widetilde{Z} \mid P_T^+) = \operatorname{Law}(\widetilde{Z} \mid P_T^-),$$
$$M(P_T^+) = M(P_T^-) = M,$$
$$\operatorname{sign}\Delta_+ = -\operatorname{sign}\Delta_- \neq 0,$$

where $\widetilde{Z} = (Z, M)$*. If* $\beta = 0$*, then* $\mathcal{C}_0$ *forces* $\gamma = 0$ *and* $\operatorname{sign}\Delta = \operatorname{sign} M$*; matched-evidence ambiguity therefore requires* $\beta > 0$*.*

Proof. Choose $0 < \delta < \min\{\beta - |M|, \frac{1}{2}\}$. For $\theta \in \{+1, -1\}$, apply Lemma 1 with the constant correctness kernel

$$\eta_\theta(x) = \tfrac{1}{2} + \theta\delta, \qquad x \in D,$$

and use the same fixed label kernel outside D. Denote the resulting target law by P_T^θ. The input marginal, predictors, score, and source law are fixed, so M is the same under P_T^+ and P_T^-. Lemma 1 also gives the same law of Z in the two worlds. Consequently, $\operatorname{Law}(\widetilde{Z} \mid P_T^+) = \operatorname{Law}(\widetilde{Z} \mid P_T^-)$ for $\widetilde{Z} = (Z, M)$.

It remains to verify class membership and the benefit signs. Under P_T^θ, $\bar{a}_\theta = \frac{1}{2} + \theta\delta$ and

$$\gamma_\theta = \theta\delta - M, \qquad |\gamma_\theta| \le \delta + |M| < \beta.$$

Thus both laws belong to $\mathcal{C}_\beta$. Lemma 2 yields $\Delta_\theta = 2\mu_T(D)\theta\delta$. The two benefits are nonzero and have opposite signs. ☐

The evidence laws coincide exactly—their total-variation distance is zero—so this is exact non-identifiability, not a rate-dependent testing bound. Randomization cannot resolve two worlds that induce exactly the same evidence law. Controlling both directional errors therefore requires abstention.

Corollary 1 (Matched-evidence abstention lower bound). *Fix* $\alpha \in (0, \frac{1}{2})$ *and the shared evidence law from Theorem 1. Let* g *be any possibly randomized label-free rule. If, in every admissible world,*

$$\mathbb{P}[g = \text{ADAPT}, \Delta \le 0] \le \alpha,$$
$$\mathbb{P}[g = \text{FREEZE}, \Delta \ge 0] \le \alpha,$$

then $\mathbb{P}[g = \text{ABSTAIN}] \ge 1 - 2\alpha$*.*

Proof. The rule has the same action probabilities in both worlds. Its ADAPT probability is at most α in the negative world. Its FREEZE probability is at most α in the positive world. The three action probabilities sum to one. ☐

4.4 Boundary Case

Inside the band, both strict signs may be feasible. At the boundary, zero benefit is feasible. These constructions differ, but both prevent a uniform strict direction.

Proposition 1 (Boundary case and closed-band abstention). *Fix* $\alpha \in (0, \frac{1}{2})$*. Under Definition 3, abstention is the pointwise maximal sound action throughout* $|M| \le \beta$*. Any label-free rule that* *controls both marginal directional errors at level α must abstain there with probability at least $1-2\alpha$. Opposite nonzero benefit signs are asserted only when $|M| < \beta$.*

Proof. For $|M| < \beta$, apply Theorem 1 and Corollary 1.

Now suppose $|M| = \beta > 0$. The declared class contains a zero-benefit world with $\gamma = -M$ and a strict-benefit world with $\gamma = 0$. These worlds have the same augmented-evidence law. In the zero-benefit world, both directional error events occur whenever the rule commits. Dual error control therefore gives $\mathbb{P}[\text{ADAPT}] \le \alpha$ and $\mathbb{P}[\text{FREEZE}] \le \alpha$. It follows that $\mathbb{P}[\text{ABSTAIN}] \ge 1 - 2\alpha$, and neither strict action is uniformly sound.

Finally, if $M = \beta = 0$, then $\mathcal{C}_0$ forces $\gamma = 0$ and $\Delta \equiv 0$. The same conclusion follows directly. □

This convention is stricter than task regret. When $\Delta = 0$, adapt and freeze are risk-equivalent, but neither certifies a strict sign.

4.5 Exact Strict-Commitment Frontier

We can now answer the population question completely. The theorem separates four jobs: reduction, sufficiency outside the band, necessity inside the band, and maximality of the resulting rule.

Theorem 2 (Exact strict-commitment frontier)**.** *Under Assumption 1, fix $(\mu_T, P_S, f_0, f_a, s)$ and let $\beta \ge 0$. Necessity and maximality hold for $\mathcal{C}_\beta$, or for a subclass closed under the constructions in Theorem 1 and Proposition 1; otherwise only sufficiency follows without an additional richness assumption.*

(i) ***Reduction.*** *For every P_T,* $\operatorname{sign}\Delta = \operatorname{sign}(M+\gamma)$ *(Lemma 2).*

(ii) ***Sufficiency.*** *If $|M| > \beta$ and $P_T \in \mathcal{C}_\beta$, then* $\operatorname{sign}\Delta = \operatorname{sign} M$.

(iii) ***Necessity.*** *If $|M| < \beta$, two laws in $\mathcal{C}_\beta$ have the same augmented-evidence law and opposite nonzero benefit signs. If $|M| = \beta > 0$, the admissible residual $\gamma = -\operatorname{sign}(M)\beta$ gives $\Delta = 0$. If $M = \beta = 0$, then $\Delta \equiv 0$ throughout $\mathcal{C}_0$. Thus $|M| \le \beta$ supports no uniform strict direction.*

(iv) ***Strict-commitment iff.*** *At the fixed augmented-evidence law, a strict action is uniformly directionally sound if and only if $|M| > \beta$. On the closed band, abstention is the pointwise maximal sound action in the sense of Definition 3.*

The population rule $M > \beta \Rightarrow$ ADAPT, $M < -\beta \Rightarrow$ FREEZE, $|M| \le \beta \Rightarrow$ ABSTAIN is the pointwise maximal sound rule on $\mathcal{C}_\beta$.

Proof. Part (i) is Lemma 2. For (ii), suppose $M > \beta$ and $|\gamma| \le \beta$. Then $M + \gamma > 0$, so Lemma 2 gives $\Delta > 0$. The negative case is symmetric.

For (iii), Theorem 1 supplies opposite nonzero signs when $|M| < \beta$. At $|M| = \beta > 0$, the boundary law with $\gamma = -\operatorname{sign}(M)\beta$ has zero benefit. When $M = \beta = 0$, every law in $\mathcal{C}_0$ has zero benefit. These cases preclude a uniform strict direction throughout the closed band.

Part (iv) combines (ii), (iii), and Proposition 1. For a nonempty compatible subclass, sufficiency gives the unique sound strict direction outside the band. This individual directional uniqueness requires a realized member: on an empty subclass, both universal directional statements are vacuous. The full compatible class is nonempty. The impossibility and boundary results rule out both strict directions on the band. □

Outside the band, the residual cannot reverse M; on the band, the same evidence admits a failed strict claim, so abstention is part of the identified answer. Let $d = \mu_T(D) > 0$. The full class identifies

$$\Delta(\mathcal{C}_\beta) = 2d\big[\max\{-\tfrac{1}{2}, M-\beta\},\\ \min\{\tfrac{1}{2}, M+\beta\}\big].$$

The bracket equals $[-\frac{1}{2}, \frac{1}{2}] \cap [M-\beta, M+\beta]$, and constant correctness kernels attain every point. A strict direction exists exactly when this interval excludes zero.

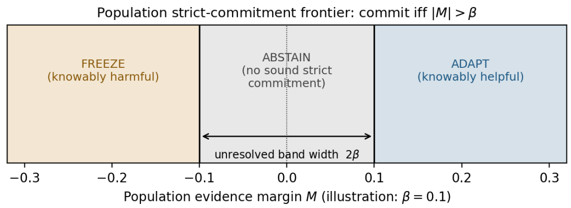

Figure 2: Population strict-commitment frontier for a fixed binary model pair under 0/1 loss. ADAPT is uniformly supported when $M > \beta$, FREEZE when $M < -\beta$, and neither strict direction is supported on $|M| \leq \beta$. This population result is not KGA's numerical cell-level rule.

Region	Meaning	Action
$M > \beta$	strictly positive benefit	ADAPT
$M < -\beta$	strictly negative benefit	FREEZE
$\|M\| < \beta$	both strict signs are feasible	ABSTAIN
$\|M\| = \beta > 0$	zero benefit is feasible	ABSTAIN
$M = \beta = 0$	benefit is identically zero	ABSTAIN

4.6 Interpretation and Limits of the Frontier

The theorem is exact only for the declared class. Choosing $\beta = 0$ assumes a zero target disagreement-conditional residual. It is not a conservative default. Without a credible upper bound on $|\gamma|$, the frontier cannot support a robust strict action. More unlabeled samples do not close this gap because the missing information lies in the target label kernel.

Empirical abstention does not prove structural unknowability. A finite-sample interval may be wide because calibration data are scarce, prediction is poor, or transfer fails. This distinction leads to the next question: can the required population residual bound be learned from the same label-free evidence?

5 What Label-Free Evidence Cannot Learn

The population frontier becomes usable only after β has been declared. Can the same label-free evidence learn a smaller valid budget? An audit must cover every residual value left unresolved on its evidence fibre.

5.1 Audit Setup and Evidence Fibres

Let W collect everything a deployment-time audit may read: the unlabeled target batch, fixed models, score function, and derived label-free statistics. It may also contain a source or calibration sample whose law does not depend on the target label kernel. A randomized procedure returns a measurable real-valued audit $\widehat{\beta} = g(W, U) \geq 0$, where the seed U has the same distribution in every world and is independent of W in every world.

For a declared class $\mathcal{C}$ and an observable-law value z, define the evidence fibre

$$\mathcal{C}(z) = \{P \in \mathcal{C} : \mathrm{Law}_P(W) = z\}$$

and its unresolved residual radius

$$\Gamma_z(\mathcal{C}) = \sup_{P \in \mathcal{C}(z)} |\gamma(P)|. \tag{6}$$

The fibre contains target laws that the audit cannot distinguish. Its radius is the smallest uniform bound on $|\gamma|$ that this observable law can support.

5.2 Exact Fibre-Wise Audit Floor

The answer is a floor, not an estimator. Any audit that is valid throughout one evidence fibre must cover the largest residual that remains possible in that fibre.

Theorem 3 (Exact label-free audit floor)**.** *Fix a nonempty fibre $\mathcal{C}(z)$ with bounded attainable residuals $\{|\gamma(P)| : P \in \mathcal{C}(z)\}$ and $\delta \in [0, 1)$. Let $\mathbb{P}_z$ denote the common joint law of (W, U) on this fibre. Suppose the measurable nonnegative audit $\widehat{\beta}$ satisfies $\mathbb{P}_P(|\gamma(P)| \leq \widehat{\beta}) \geq 1 - \delta$ for every $P \in \mathcal{C}(z)$. Then*

$$\mathbb{P}_z\Big(\widehat{\beta} \geq \Gamma_z(\mathcal{C})\Big) \geq 1 - \delta.$$

The constant audit $\widehat{\beta} \equiv \Gamma_z(\mathcal{C})$ is valid with $\delta = 0$. Consequently, on the displayed high-probability event, an audited frontier can commit only where the frontier with radius $\Gamma_z(\mathcal{C})$ already commits.

Proof. The common joint law of (W, U) induces the same law for $\widehat{\beta}$ in every world. Choose $P_n \in \mathcal{C}(z)$ so that $|\gamma(P_n)| \uparrow \Gamma_z(\mathcal{C})$. Under the common law, each event $\{\widehat{\beta} \geq |\gamma(P_n)|\}$ has probability at least $1-\delta$. These events decrease to $\{\widehat{\beta} \geq \Gamma_z(\mathcal{C})\}$, so continuity from above proves the bound. The constant audit is valid by the definition of the supremum. Finally, $\widehat{\beta} \geq \Gamma_z$ and $|M| > \widehat{\beta}$ imply $|M| > \Gamma_z$. □

The theorem does not prohibit useful budgets. An audit using the same observable law must cover every residual left possible on that fibre. A smaller bound requires labels or structure that genuinely refines the compatible set; $\Gamma_z(\mathcal{C})$ is an oracle fibre benchmark, not a general finite-batch estimator.

Declared budget and abstention radius. For the full declared class $\mathcal{C}_\beta = \{P : |\gamma(P)| \leq \beta\}$ with $0 \leq \beta \leq \frac{1}{2}$, feasible $M \in [-\frac{1}{2}, \frac{1}{2}]$, and positive disagreement mass, the label-kernel construction in Lemma 1 gives

$$\Gamma_z(\mathcal{C}_\beta) = \beta. \tag{7}$$

Thus the least valid fibre-wide residual bound equals the population abstention radius, β. This connection does not turn the declared β into an estimate.

When calibration labels do not shrink the target fibre. A labeled calibration domain does not reduce the target fibre unless the declared class links its label kernel to deployment. Lemma 1 can otherwise vary deployment labels while leaving the calibration sample unchanged. Labels therefore help through representativeness or structural coupling, not merely through their existence.

What additional structure can buy. Extra information can shrink the fibre, but it changes the problem. The available routes are representative deployment labels, exchangeable historical deployment episodes, a structural cross-domain coupling, or a domain-specific physical bound. Each gain comes from labels or assumptions, not the current unlabeled batch.

5.3 Implications for KGA

KGA does not estimate M, γ, or β, and does not apply $|M| > \beta$ numerically. It predicts Δ^{cell} and uses an empirical radius ε. Transferring cell-level coverage to population benefit requires an additional sampling-error argument that the reported experiments do not provide. The next section therefore changes the question from population identification to deployment-time prediction.

6 KGA: An Empirical Benefit-Interval Gate

For the current evaluation unit, can KGA support which side of zero the benefit lies on? It predicts cell benefit from label-free evidence and attaches a residual-calibrated radius.

6.1 Shared Decision Logic, Different Targets

Scope of KGA. KGA predicts measured evaluation-cell benefit Δ^{cell}, not population benefit Δ. Its radius ε describes empirical prediction error under a named calibration protocol. It neither estimates the population residual bound β nor establishes population-risk coverage.

Both layers commit only when their respective uncertainty set supports one strict sign, but they use different targets:

$$\begin{gathered} \Delta = 2\mu_T(D)(M + \gamma), \\ \Delta^{\text{cell}} = S(f_a; \mathcal{E}) - S(f_0; \mathcal{E}), \qquad \widehat{\Delta} = h_\theta(Z). \end{gathered} \tag{8}$$

Development cells fit h_θ, and disjoint residual-calibration cells supply residuals. A scored cell contributes Z, but not its outcome. The empirical coverage premise is

$$\mathbb{P}(|\widehat{\Delta} - \Delta^{\text{cell}}| \leq \varepsilon) \geq 1 - \alpha. \tag{9}$$

Proposition 2 applies only when this premise is justified.

6.2 The Coverage-to-Action Proposition

Proposition 2 (Coverage-to-action implication)**.** *Fix the scalar benefit target and its evaluation unit before calibration. Let B and $\widehat{B}$ be finite real random variables, and let $\varepsilon \in [0, \infty]$ be a measurable random radius on a common probability space containing all calibration, benefit-model fitting, candidate-construction, and evaluation-unit randomness. The prediction $\widehat{B}$ and radius ε must be available without reading the new unit's evaluation labels. For $\alpha \in (0, 1)$, suppose the interval satisfies marginal coverage*

$$\mathbb{P}\big[\,|\widehat{B} - B| \le \varepsilon\,\big] \;\ge\; 1 - \alpha.$$

Define

$$g = \begin{cases} \textsc{adapt} & \widehat{B} - \varepsilon > 0, \\ \textsc{freeze} & \widehat{B} + \varepsilon < 0, \\ \textsc{abstain} & \textit{otherwise.} \end{cases}$$

Then, marginally over that common probability space,

$$\mathbb{P}[g = \textsc{adapt}, B \le 0] \le \alpha,$$
$$\mathbb{P}[g = \textsc{freeze}, B \ge 0] \le \alpha.$$

Proof. On the event $\{|\widehat{B} - B| \le \varepsilon\}$, the condition $\widehat{B} - \varepsilon > 0$ implies $B > 0$, so a false adapt requires the complement of this event, which has probability at most α. The freeze case is symmetric. □

For the empirical panels, $B = \Delta^{\mathrm{cell}}$ and $\widehat{B} = \widehat{\Delta}$. A population-risk application instead sets $B = \Delta$ and requires coverage for that quantity; cell-outcome coverage alone does not provide it.

Remark 2 (Scope of the implication)*.* Proposition 2 controls marginal error for the declared target. It does not control error conditional on commitment, group-specific coverage, candidate-selection multiplicity, or repeated-time behavior. The coverage premise needs separate justification. When it is unsupported, a declared fallback may retain f_0 and record ABSTAIN, but that response does not prove f_0 safe or optimal and is not a certified FREEZE decision. For this certificate, a population sign claim requires interval coverage for population benefit itself, or a justified cell-to-population transfer bound. Once that coverage premise holds, the event-containment argument applies directly; it does not need an additional risk-alignment assumption.

6.3 Cell-to-Population Transfer

Cell coverage alone does not imply population coverage. The following proposition states the additional sampling premise needed to change the target while preserving the three-way decision semantics. It is a conditional extension, not a replacement of KGA's cell-level calibration. The evaluation unit, target-environment population, model-pair construction protocol, score functional, and units must be declared before calibration.

Proposition 3 (Cell-to-population transfer)**.** *Let $\widehat{\Delta}$, Δ^{cell}, and Δ be real-valued measurable quantities on a common probability space, with cell and population benefit defined for the same model pair and score functional. Let $\varepsilon, b \ge 0$ be finite measurable radii. Suppose that $\alpha_{\mathrm{cell}}, \delta \in [0, 1)$, $\alpha_{\mathrm{cell}} + \delta < 1$, and*

$$\begin{aligned} &\mathbb{P}\Big(|\widehat{\Delta} - \Delta^{\mathrm{cell}}| \le \varepsilon\Big) \ge 1 - \alpha_{\mathrm{cell}}, \\ &\mathbb{P}\Big(|\Delta^{\mathrm{cell}} - \Delta| \le b\Big) \ge 1 - \delta. \end{aligned} \tag{10}$$

Then, without requiring independence of these two events,

$$\mathbb{P}\Big(|\widehat{\Delta} - \Delta| \le \varepsilon + b\Big) \ge 1 - \alpha_{\mathrm{cell}} - \delta. \tag{11}$$

If $\widehat{\Delta}$, ε, and b are available before the new evaluation labels, define $L_{\mathrm{pop}} = \widehat{\Delta} - (\varepsilon + b)$ and $U_{\mathrm{pop}} = \widehat{\Delta} + (\varepsilon + b)$. The rule

$$g_{\mathrm{pop}} = \begin{cases} \textsc{adapt}, & L_{\mathrm{pop}} > 0, \\ \textsc{freeze}, & U_{\mathrm{pop}} < 0, \\ \textsc{abstain}, & \textit{otherwise} \end{cases} \tag{12}$$

satisfies

$$\begin{aligned} \mathcal{E}_{\mathrm{pop}} = \{&g_{\mathrm{pop}} = \textsc{adapt},\ \Delta \le 0\} \\ &\cup \{g_{\mathrm{pop}} = \textsc{freeze},\ \Delta \ge 0\}, \end{aligned} \tag{13}$$
$$\mathbb{P}(\mathcal{E}_{\mathrm{pop}}) \le \alpha_{\mathrm{cell}} + \delta.$$

Proof. On the intersection of the two premise events, the triangle inequality gives $|\widehat{\Delta} - \Delta| \le |\widehat{\Delta} - \Delta^{\text{cell}}| + |\Delta^{\text{cell}} - \Delta| \le \varepsilon + b$. The complement of that intersection has probability at most $\alpha_{\text{cell}} + \delta$ by the union bound. On the intersection, $L_{\text{pop}} > 0$ implies $\Delta > 0$ and $U_{\text{pop}} < 0$ implies $\Delta < 0$, so either false strict direction requires failure of at least one premise event. □

The bound is marginal over the declared evaluation unit and its protocol randomness, not conditional on commitment, groupwise, or anytime-valid. The radii need separately justified coverage premises; computing their sum does not establish those premises. No such sampling-error bound is established for the reported panels, whose gates continue to use ε, not $\varepsilon + b$. An unavailable premise or non-finite radius therefore supports no population certificate: the operational fallback is ABSTAIN while retaining f_0, not a certified FREEZE. The sampling radius b is distinct from the population residual bound β. Appendix A records the sampling and score-functional restrictions.

Algorithm 1 KGA decision for one proposed candidate

Require: Frozen f_0, adapter $\mathcal{A}$, unlabeled X, frozen h_θ and calibration artifacts

Create $f_a \leftarrow \mathcal{A}(f_0, X)$ in isolated shadow state
Extract $Z = \phi(X, f_0, f_a)$ and validate the assessment inputs
if the assessment is unavailable **then**
    retain f_0, log the reason, and return ABSTAIN
end if
Obtain ε from the calibrated residual rule in Eq. (14)
$\widehat{\Delta} \leftarrow h_\theta(Z)$
$L \leftarrow \widehat{\Delta} - \varepsilon$; $U \leftarrow \widehat{\Delta} + \varepsilon$
if $L > 0$ **then**
    commit f_a and return ADAPT
else if $U < 0$ **then**
    discard f_a and return FREEZE
else
    retain f_0, log uncertainty, and return ABSTAIN
end if

6.4 Evidence, Calibration, and the Action Rule

Each candidate-specific model uses label-free summaries of confidence, class balance, model disagreement, distribution shift, and update magnitude. Natural tracks retain protocol-specific models and schemas. The evidence inventory in Appendix I records their exact specifications.

Given residuals $r_i = |\widehat{\Delta}_i - \Delta_i^{\text{cell}}|$, ordered as $r_{(1)} \le \cdots \le r_{(n)}$, the order-statistic rule is

$$k = \lceil (n+1)(1-\alpha) \rceil, \qquad \varepsilon = \begin{cases} r_{(k)}, & k \le n, \\ +\infty, & k > n. \end{cases} \tag{14}$$

The rank rule has split-conformal coverage when calibration and next-unit scores are jointly exchangeable under the full fitting and candidate-construction protocol (Tibshirani et al., 2019; Barber et al., 2023); here n counts cells, not images. Disjoint fit, calibration, and check labels prevent the specified reuse; this does not establish exchangeability. The painting-development inclusion rates below are descriptive checks on opened cells, not validation of group-conditional, repeated-use, or population-risk coverage.

A deterministic five-fold cross-fit is keyed by stable cell IDs. For each score fold, its complement is divided into disjoint estimator-fit and residual-calibration subsets. Fit outcomes determine the predictor, and the exact-rank rule on calibration residuals determines the radius. No scored outcome can affect its own prediction, radius, or action; the whole-pipeline invariance test verifies this property. This separation blocks direct outcome flow but does not create independent environments or establish exchangeability. Natural tracks use separate partitions only where their archived protocols record them.

For radius 0.01, predictions 0.04, −0.04, and 0.005 produce ADAPT, FREEZE, and ABSTAIN, respectively. These values illustrate the rule; they are not experimental results.

Runtime includes shadow adaptation, two-model inference, evidence extraction, and storage; comparable end-to-end latency measurements are unavailable.

7 Primary Experimental Protocols

7.1 Two Primary Questions and the Evidence Hierarchy

The CIFAR-10-C diagnostic asks whether KGA distinguishes helpful from harmful updates on an opened constructed grid. CCT-20 asks whether a frozen gate avoids measured candidate harm across camera locations. The first exercises directional branches retrospectively; the second evaluates governed retention and may pass without ADAPT exposure.

The opened, dependent, constructed CIFAR-10-C diagnostic (Hendrycks and Dietterich, 2019) uses six corruption families at severities 1 and 5. It crosses three batch sizes (200, 16, and 8), three compositions (IID, imbalanced, and single-class), two update schedules (10 steps at 10^{-3} and 50 at 2.5×10^{-3}), and two repeats: 432 cells per RNG seed and 2,160 per candidate over seeds 0–4.

Tent, EATA, and SAR are local ports applied to one ResNet-18 checkpoint; the five seeds repeat streams and adapter randomness, not checkpoint training. Point estimates weight cells equally; family sensitivity first averages within each family and then weights the six means equally.

The recorded candidates are local, protocol-matched composite operating points rather than untouched reproductions of published defaults. They combine evaluation-batch BatchNorm statistics with normalization-affine updates; Appendix D records the contract and unexecuted BatchNorm-only control.

The three-way cross-fit removes each scored fold's outcomes from fitting and residual calibration, but does not make the shared checkpoint, corpus, or cells independent. This is a retrospective diagnostic, not prospective confirmation or an untouched deployment trial.

7.2 CCT-20 Design and Locked Utility-Preservation Endpoint

The CCT-20 study (Beery et al., 2018) crosses 5 independent checkpoints with 9 target locations, giving 45 cells and 6950/16325 probe/evaluation images (23275 total). Set-membership top-1 accuracy credits any archived category, not an official single-label target. Official Tent makes one non-episodic Adam update per batch at 10^{-3}; pre-update logits score it, state carries within a cell, and resets between cells.

Individual outcomes remained unopened until one-shot scoring, but aggregate target-label metadata had previously been inspected; CCT-20 was therefore cell-outcome-unopened and internally sealed, not globally label-unopened or publicly preregistered. Ridge fitting used 10 rows from two camera locations and calibration used 45 rows across nine others; transfer to target locations is assumed.

For fixed policy b, G_b is mean baseline regret minus mean KGA regret. The limited, predeclared utility check uses nominal pointwise 95% intervals and passes when their lower endpoints satisfy

$$L_{\text{adapt}} > 0, \qquad L_{\text{freeze}} > -0.005. \tag{15}$$

The second threshold is frozen-policy noninferiority; this check can pass when KGA always serves f_0 and is not a population guarantee.

The separate strong-success rule requires complete data for all 45 cells, nominal 97.5% Bonferroni intervals above zero against both fixed policies, both exact nine-location tests surviving Holm at 0.05, ADAPT and FREEZE rates at least 0.10, and total commitment at least 0.30. It therefore requires interval-supported improvement over both fixed policies; any failure precludes strong success.

The CCT utility table 6 uses pointwise 95% intervals; the separate strong-success audit 20 uses 97.5% Bonferroni intervals. They are not alternate versions of the same interval.

Table 4: Evidence status and admissible claim. "Primary" means central to the paper, not prospectively confirmatory; unfavorable and incomplete studies remain in scope.

Evidence	Design and status	Question it can answer
CIFAR-10-C	opened, dependent, constructed; cell-outcome-disjoint cross-fit	retrospective candidate-specific routing
CCT-20	cell-outcome-unopened, internally sealed execution; aggregate target metadata previously inspected	conservative retention under the locked endpoint
So2Sat-LCZ42	development stop before target access	candidate failed eligibility
DomainNet painting	opened development; five checkpoints on shared cells and separate reference-source candidates; fixed follow-up diagnostics	candidate feasibility, abstention, and mechanism limits; not confirmation
Auxiliary replays	opened or one-sided diagnostics	transfer limits and supporting behavior
Invalid or partial records	PACS partial; iWildCam metric invalid; historical ports superseded; Office-Home serialization invalid	audit history only

Scoring and inference. Actions are fixed before outcomes are joined, but candidate TTA is transductive because it reads unlabeled evaluation inputs. CIFAR sensitivity resamples the six corruption-family means conditional on the archived checkpoint. CCT-20 uses a paired product bootstrap over checkpoint rows and location columns; its separate exact sign-flip reference calculation uses nine location means and requires a stronger joint sign-symmetry condition than exchangeability alone. All reported levels are nominal. The inference audit in Appendix F gives the metrics, sign convention, full assumptions, and retrospective status (Hemerik and Goeman, 2018).

8 Primary Results

8.1 CIFAR-10-C: Directional Actions and Candidate-Specific Utility

Table 5 first exposes the ADAPT/FREEZE/ABSTAIN record under one current rule. Tent makes two false ADAPT decisions and one false FREEZE; EATA makes neither; SAR makes no FREEZE decision, so that error direction is unexercised. These counts require their action and all-cell denominators below. Utility is a secondary measured consequence: Tent and EATA have lower point regret than both fixed policies, whereas always-adapt has lower regret than KGA for SAR. The panel therefore supports only candidate-specific, retrospective action and utility results.

Tent regret is 0.0019, compared with 0.0080 for always-adapt and 0.1239 for always-freeze. EATA's values are 0.0016, 0.0033, and 0.1313. Their ADAPT/FREEZE/ABSTAIN counts are 1,091/337/732 and 1,207/105/848, respectively. Tent makes two false ADAPT decisions; EATA makes none. SAR makes 1,414/0/746 decisions. Its KGA regret is 0.0018, compared with 0.0003 for always-adapt and 0.1405 for always-freeze. Abstention can prevent harmful updates while still forgoing helpful ones.

The nominal six-family intervals distinguish Tent from EATA. Gaps are baseline regret minus KGA regret, so positive values favor KGA. Tent's gap versus always-adapt is 0.0061, with nominal family-bootstrap interval $[0.0021, 0.0095]$; its interval versus always-freeze is also strictly positive. EATA's interval versus always-adapt crosses zero, $[-0.0013, 0.0044]$, although its interval versus always-freeze is positive. Tent therefore has the favorable nominal two-baseline pattern; EATA does not.

The retrospective reference tests do not make either pattern confirmatory. Across the six candidate-by-baseline contrasts in the retrospective comparison, Tent's Holm values are 0.140625 against always-adapt and 0.09375 against always-freeze. EATA's are 0.3125 and 0.09375, respectively. The replay, family unit, tests, and multi-

Table 5: Opened, dependent CIFAR-10-C diagnostic over 2,160 equally weighted cells per candidate. Lower regret is better. A/F/U gives action exposure; directional errors use the corresponding action denominator. Tent's all-cell errors are 2/2,160 for false ADAPT and 1/2,160 for false FREEZE. All cells reuse one checkpoint, so this is retrospective rather than independent environment validation. Canonical panel SHA-256: `35d4c165843de1ece3cb35ffb4ac50dbfcbfa33b646754e2b6d133fbdfa78c6e`.

Candidate	n	KGA	Adapt	Freeze	A/F/U	False A/A	False F/F	Commitment
Tent	2160	0.0019	0.0080	0.1239	1091/337/732	2/1091	1/337	0.6611
EATA	2160	0.0016	0.0033	0.1313	1207/105/848	0/1207	0/105	0.6074
SAR	2160	0.0018	0.0003	0.1405	1414/0/746	0/1414	–	0.6546

plicity analysis were selected after the grid was open, and all six families share one checkpoint. Table 16 reports every interval and reference value. Nominal interval exclusion and retrospective Holm adjustment answer different questions; neither establishes prospective coverage.

8.2 Retrospective Interval Diagnostics

Retrospective inclusion is 1,976/2,160 for Tent, 1,950/2,160 for EATA, and 1,949/2,160 for SAR. These values describe opened cross-fit cells; they are not independent validation of coverage on new environments. The inference appendix F reports interval widths and family-level diagnostics. Tent's two false ADAPT actions correspond to marginal error 2/2160 and conditional error 2/1091. It also makes one false FREEZE among 337 FREEZE decisions. EATA makes no false ADAPT and no false FREEZE. SAR makes no false ADAPT and no FREEZE decisions, so its conditional false-freeze rate is undefined. Exposure and denominators are essential: a zero count alone is not evidence of safety.

8.3 CCT-20: Conservative Retention, Not Selective Routing

All 45 checkpoint–location cells completed. KGA made 0/44/1 ADAPT/FREEZE/ABSTAIN decisions. All 45 served predictions used the frozen model, but the sole helpful cell received FREEZE, yielding one false FREEZE among 44 FREEZE actions (1/45 overall). Tent helped in 1 cell, tied in 0, and harmed in 44; 43 harmful cells received FREEZE, and the remaining harmful cell received ABSTAIN. The study therefore passes only the locked utility-preservation endpoint and does not demonstrate selective routing or strong success.

KGA matched always-freeze and avoided the measured degradation from always-adapt. The unchanged locked rule therefore returns pass on the stored cells (generated flag: yes). The archived verdict token remains SAFE UTILITY ONLY, but the supported description is utility preservation only. This is a scoped endpoint, not a finite-sample population-safety guarantee.

The complete strong-success audit is more informative. Completeness passed. Against always-adapt, the nominal 97.5% Bonferroni interval is $[0.08501071, 0.31066374]$ and the Holm-adjusted location sign-flip value is 0.00390625; both pass their locked thresholds. Against always-freeze, the contrast and interval are zero and the Holm value is 1; neither supports strict improvement. FREEZE exposure and total commitment exceed 0.10 and 0.30, but ADAPT exposure is zero and fails its 0.10 threshold. The strong-success criterion therefore fails. This is conservative frozen-model retention, not learned selective routing. The false-adapt count (0/45) is mechanically zero because ADAPT exposure is zero.

Every development cell was harmful for Tent, so the fit saw no helpful example. Only two camera locations supplied fitting data, and calibration-to-target transfer is unverified. The complete location ledger and primary inference appear in the CCT-20 ledger in Appendix G.1.

Table 6: CCT-20 action record and locked utility-preservation endpoint over 45 checkpoint–location cells. Positive policy contrasts favor KGA. The 95% intervals belong to the utility endpoint; Appendix G reports the separate 97.5% Bonferroni strong-success audit.

Item	Result
Actions	0 ADAPT / 44 FREEZE / 1 ABSTAIN
Measured effects	1 helpful / 0 tied / 44 harmful
False FREEZE	1/44 conditional; 1/45 overall
Locked endpoint	Pass: utility preservation only
Strong success	Fail: no ADAPT exposure or improvement over always-freeze

Comparator	Mean contrast	Nominal 95% CI	Required lower bound
Always adapt	0.1819	[0.0937, 0.2914]	$L > 0.0000$
Always freeze	0.0000	[0.0000, 0.0000]	$L > -0.0050$

We infer no selective-routing success from this result.

8.4 Development Stops and Adverse Diagnostics

So2Sat-LCZ42 (Zhu et al., 2020) screened Tent and SAR in a development-only, multi-city, multi-checkpoint feasibility study. Tent produced both helpful and harmful development cases, but neither candidate passed every predeclared eligibility check. The study therefore stopped before gate calibration. No gate-calibration rows or target inputs, pixels, or labels were accessed, so there is no target natural-shift score. The gate-fit outcomes are now open; retuning them cannot create fresh confirmation.

DomainNet painting development. Five separately trained checkpoints were evaluated on the same 24 check cells per checkpoint. KGA abstained on all 120 seed-cell records; candidate effects were helpful in 12, harmful in 15, and tied in 93. Equal-seed regret was 0.110677 percentage points for both KGA and always-freeze, versus 0.123547 for always-adapt. These repetitions do not create 120 independent environments. Using a separate reference-source ResNet-50 checkpoint, a one-pass local entropy candidate restored the source BatchNorm running buffers before inference. Its 24-cell check panel had 7 helpful, 14 harmful, and 3 tied cells. A separate AdaContrast-derived single-process recipe had 1 helpful and 23 harmful cells. Both gates abstained on all 24 cells. The two recipes produced different frozen prediction totals and are not a matched adapter comparison. These are opened-development negatives, not selective-routing or unseen-shift coverage evidence. Fixed queue and normalization-state diagnostics did not rescue the evaluated candidates; their scope and separate numerical summaries appear in the DomainNet development appendix B.11.

The auxiliary studies expose further limits. ImageNet-R and PACS favor always-adapt, while ImageNet-C depends strongly on candidate and operating point. Office-Home and RxRx1 mainly retain the frozen model; the opened Camelyon17 OOD panel adapts throughout. The auxiliary-results appendix B reports valid diagnostic rows on separate metric scales. It also explains why iWildCam, the historical POEM/AETTA ports, and the focused Office-Home route cannot support current headline comparisons.

ImageNet-C SAR operating point. The ImageNet-C SAR result uses a local port at learning rate 4×10^{-3}, which is $16\times$ the published 2.5×10^{-4} setting. The port adapts the recorded BatchNorm affine parameters, including the upper ResNet block that the official default excludes. It also combines target-batch BatchNorm-statistics replacement with gradient updates. This is an aggressive local diagnostic, not an official paper-default configuration. We ran no matched control at SAR's published operating point, so the result concerns this port and cannot be read as evidence against SAR as published.

ImageNet-R backbone pattern. KGA has higher regret than always-adapt on 8 of 10 backbones. On `efficientnet_b0`, KGA improves over always-adapt but ties always-freeze at zero regret. On `swin_t`, it also improves over always-adapt but ties always-freeze at 0.00473958 regret. Thus KGA has lower regret than both fixed policies on 0 of 10 backbones. Four of the eight adapt-favoring backbones have a 0% harmful-update base rate, so routing cannot improve on always-adapt there. This opened, descriptive pattern is consistent with routing being most useful when harmful updates occur. It is not one deployable multi-backbone policy or confidence-supported natural-shift evidence.

9 Limitations and Broader Impact

The measured action record does not demonstrate universal safety. The theory is exact relative to its declared class, whereas the empirical intervals are protocol-scoped and require separate coverage assumptions. The recorded experiments measure directional errors and utility only for the fixed candidates, metrics, and evaluation units studied. Neither layer establishes unconditional deployment safety.

9.1 Formal Assumptions and Empirical Calibration Scope

The population frontier is exact within its declared rich binary class. A credible β must come from domain knowledge, target labels, or a stated transfer assumption. A narrower class may permit stronger decisions. An unsupported bound cannot justify a strict population claim. The substantive object is the class-dependent feasible set, not the inequality by itself.

The development-based surrogate failed. We tested estimating β from source-like development cells rather than declaring it. The estimate is 1.4–50× too small on real evaluation cells (24–73% of cells fall outside the declared class $\mathcal{C}_\beta$; commit error 0.4–16.6%), and returns zero commitments on all 405 ImageNet-C cells in 5 of 10 configurations. We therefore withdraw the estimation route and declare β from domain knowledge instead.

This test used an operational development proxy; it could not observe the theorem's target disagreement-conditional residual directly. The repository retains the reported result text but not the named raw beta-sweep artifact. The negative result is therefore historically documented but not presently regenerable from the canonical bundle. KGA neither estimates nor numerically uses β.

For KGA, empirical interval coverage needs separate support at the intended decision unit. A larger ε reduces commitment but may increase regret when f_0 is worse. A smaller ε increases exposure and the cost of calibration failure. Changing the adapter, checkpoint, feature schema, or update settings changes the prediction problem. Each change requires recalibration or a justified transfer argument.

Repeated deployment needs a separate guarantee. A predeclared policy does not by itself create exchangeability or time-uniform safety. Reusing calibration data, replacing the frozen baseline after an accepted update, or selecting later candidates from earlier outcomes changes the joint process. Adaptive conformal inference studies online coverage frequency with feedback (Gibbs and Candes, 2021); it does not automatically give time-uniform control of KGA's false commitments. Selection-conditional and group-conditional guarantees require additional methods, as Section 2 explains.

9.2 Evidence Needed for Prospective Confirmation

The controlled results reuse one checkpoint, one corpus, and related corruption cells. Their cell-outcome-disjoint split fixes a leakage defect; it does not remove dependence. The interval inclusion table is retrospective, not an independent calibration test. The natural/development record remains retention-heavy or adverse: CCT-20 has no ADAPT exposure, So2Sat stops during development, and the painting-development gates abstain throughout. Five-checkpoint repetition documents development behavior but supplies neither independent unseen environments nor a prospective routing success. Empirical abstention may reflect weak prediction, limited data, a wide ε, or transfer failure. It does not establish population non-identifiability.

The search history is substantial. One superseded census reported 1,387 determinations and 326 positives (23.5%). A broader archived scan reported 1,427 determinations and 345 positives (24.2%). Neither census froze a reproducible family definition, file list, and repository state. They are overlapping historical search summaries, not two predeclared families and not the multiplicity family for the current panel. The controlled results remain explicitly post hoc.

Table 7 states the four missing tests and the claim each would inform. They cannot be supplied by reclassifying the existing panel.

The confirmation protocol in Appendix K specifies a future protocol.

9.3 Operational Risks and Deployment Scope

Rejecting an update can prevent adaptation harm. A frozen fallback can prevent adaptation-induced harm while still preserving a model that is inaccurate, biased, or unsuitable for the application. Treating abstention as a safety certificate would be especially risky. A deployment should retain the assessment reason and route unresolved cases to domain-appropriate review or evidence acquisition. Our experiments do not validate clinical, safety-critical, or autonomous operation. We claim neither a held-out physical camera deployment nor a common end-to-end latency result.

The maintained software distinguishes an unavailable assessment from a supported negative interval. Its tests and proof artifacts document a scoped validity, commitment-control, and rollback layer. They do not certify every data pipeline or production system. The appendices record reproduction instructions, formal scope, excluded historical extensions, and release checks.

10 Conclusion

K-Bound separates missing target-label information from estimable uncertainty. For a fixed binary model pair under 0/1 loss, a declared residual class induces a feasible set of population benefits. That set supports a strict direction exactly outside the closed band $|M| \leq \beta$. The evidence-fibre audit theorem explains why additional functions of the same label-free evidence cannot remove residual ambiguity that remains on the fibre.

KGA addresses a different target. It predicts measured evaluation-cell benefit and uses a separately calibrated interval to ADAPT, FREEZE, or ABSTAIN. Its empirical criterion is directional action correctness, with exposure and false-action denominators reported separately. The utility preservation endpoint is reported separately, not as proof of universal safety. The current CIFAR-10-C diagnostic exercises both directional actions retrospectively, whereas CCT-20 demonstrates conservative all-frozen retention rather than selective routing. These results support evidence-aware commitment under recorded protocols, not unconditional safety or population transfer. The newer opened development record likewise shows abstention and unrecovered candidate harm.

The practical lesson is simple: commit a proposed update only when the relevant target, evidence, and assumptions support a direction; otherwise preserve the fallback and record the unresolved uncertainty.

Table 7: Evidence needed for prospective confirmation. Each row names a missing test rather than reinterpreting a completed diagnostic.

Missing test	Question it would answer	Present boundary
Synchronized official comparators	Does the current policy add value beyond official TTA/ALine, AETTA, POEM, and simple benefit gates?	Only historical local-port diagnostics exist.
New natural shift with helpful and harmful candidates	Can a frozen policy exercise nonzero ADAPT and FREEZE exposure prospectively?	Current natural tracks are one-sided, retention-heavy, or stopped.
Untouched-group coverage evaluation	Do interval inclusion and directional errors transfer beyond opened cross-fit cells?	No held-out group-coverage result is available.
Independent checkpoints and environments	Does the inference extend beyond one checkpoint and related constructed conditions?	Five separately trained checkpoints have development results on shared painting cells; independent-environment prospective inference remains unavailable. The primary CIFAR grid reuses one checkpoint and corpus.

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A Conditional Bridge from Cell Coverage to Population Coverage

The interval proposition covers its declared target. Proposition 3 now states the conditional bridge explicitly. This appendix records what its additional premise requires and what it does not establish for the reported experiments.

Additional condition for population transfer. For cell outcomes using the same loss as the population risk, such as accuracy and 0/1 loss, suppose that, on the same probability space,

$$\mathbb{P}(|\widehat{\Delta} - \Delta^{\text{cell}}| \leq \varepsilon) \geq 1 - \alpha_{\text{cell}},$$
$$\mathbb{P}(|\Delta^{\text{cell}} - \Delta| \leq b) \geq 1 - \delta,$$

where $b \geq 0$ is available without the new evaluation labels, then

$$\mathbb{P}(|\widehat{\Delta} - \Delta| \leq \varepsilon + b) \geq 1 - \alpha_{\text{cell}} - \delta.$$

As in the main-paper proposition, the triangle inequality applies on the intersection of the two coverage events, and the union bound controls its complement; independence is not needed. A population interval-supported action would use radius $\varepsilon + b$ and budget $\alpha_{\text{cell}} + \delta \leq \alpha_{\text{target}}$. No such sampling-error bound is established for the reported panels, and their gates do not apply this enlarged radius. This conditional implication therefore adds no new population-safety claim to the empirical results.

For paired accuracy on fresh IID evaluation observations, a direct fixed-pair IID concentration argument requires the model pair to be fixed relative to that evaluation sample, under a stated sampling model. Merely hiding evaluation labels while adapting on the evaluation inputs does not establish this premise. Likewise, disjoint windows by themselves do not imply IID sampling or independence across environments. The reported transductive panels are not retroactively assigned a sampling radius.

The maintained Lean namespace `KBound.PopulationTransfer` verifies the triangle, union-bound, and strict-direction implications under their declared premises. Formal verification does not verify that the empirical calibration or sampling premises hold for a dataset.

For a nonlinear score such as macro-F1, an analogous argument must first define the population functional and a suitable sampling-error bound. Neither follows from the binary disagreement identity. Appendix L gives a counterexample with perfect cell-outcome coverage and negative population benefit.

B Additional Results and Non-Promoted Studies

This section preserves evidence outside the two primary protocols. Ordinary accuracy and balanced accuracy remain separate, and no result is pooled across datasets, candidates, or protocols. The input inventory is bound by source-manifest SHA-256 `03b1d2b1e9e5ed1cf835126f871ed83eb8497a1ac817 27b3955af0d69eb89742`. Historical, incomplete, and invalid studies retain their original status.

B.1 Accuracy-Based Diagnostic Replays

Status: retrospective diagnostic.

Table 8 collects the remaining ordinary-accuracy replays; the dataset-specific interpretations follow below.

Table 8: Appendix ordinary-accuracy diagnostics. Each row is weighted over the evaluation unit named in the table; lower regret is better. Commitment counts ADAPT or FREEZE and is not interval coverage. The rows are retrospective or historical, not confirmatory. Office-Home's primary and stream-seed rows differ from its invalid five-checkpoint audit. ImageNet-C is candidate-specific. For PACS, $n = 12$ denotes equal-weight domain–seed regret units, whereas the archive contains 216 repeated action decisions.

Protocol	Status	n	KGA	Adapt	Freeze	FA_u	Commitment rate
Office-Home primary	retrospective retention	35	0.0158	0.0468	0.0158	0.0000	0.3143
Office-Home stream seeds	descriptive replication	54	0.0215	0.0458	0.0217	0.0000	0.2778
ImageNet-C Tent	candidate-specific	135	0.0154	0.0191	0.0145	0.0222	0.0370
ImageNet-C EATA	candidate-specific	135	0.0024	0.0001	0.0342	0.0074	0.8074
ImageNet-C SAR	candidate-specific	135	0.0227	0.0529	0.0319	0.0000	0.2741
PACS (aggregate)	partial diagnostic	12	0.0431	0.0176	0.0446	0.0093	0.8148
CIFAR-10.1	locked diagnostic	48	0.0017	0.0190	0.0017	0.0000	0.4583

B.2 Balanced-Accuracy Diagnostic Replays

Status: retrospective diagnostic on a separate metric scale.

Table 9 reports the balanced-accuracy replays on their own metric scale, with each recorded candidate and evaluation unit kept distinct.

Table 9: Appendix balanced-accuracy diagnostics; lower regret is better. Each row uses the evaluation-cell weighting recorded by its protocol. The archived contracts name balanced accuracy for Camelyon17, ImageNet-R, and RxRx1; equality with ordinary accuracy can occur only under recorded class balance. ImageNet-R averages candidate backbones rather than one deployed gate. Camelyon17 OOD was opened, its B–v2 rows are separate within-seed diagnostics, and the displayed RxRx1 row is model seed 0. No interval or confirmatory claim is attached to this table.

Protocol	Status	n	KGA	Adapt	Freeze	FA_u	Commitment rate
ImageNet-R backbones	architecture panel	480	0.0137	0.0064	0.0325	0.0125	0.4437
Camelyon17 OOD	opened one-sided	18	0.0000	0.0000	0.1381	0.0000	1.0000
Camelyon17 B–v2 Tent	opened diagnostic	108	0.0446	0.0097	0.0820	0.0000	0.2500
Camelyon17 B–v2 EATA	opened diagnostic	108	0.0260	0.0040	0.0911	0.0278	0.4722
Camelyon17 B–v2 SAR	opened diagnostic	108	0.0289	0.0016	0.1001	0.0093	0.4815
RxRx1 model seed 0	retention diagnostic	60	0.0000	0.2531	0.0000	0.0000	1.0000

B.3 Large-Scale Corruption: ImageNet-C

Status: retrospective candidate-specific diagnostic.

ImageNet-C is candidate-dependent. At the regenerated cross-fit authority, SAR has KGA/adapt/freeze regrets 0.0227/0.0529/0.0319 and 28/9/98 A/F/U decisions. It has a point advantage over both fixed policies and no false adaptation among 135 cells, but no seed-robust inferential support. Tent has regrets 0.0154/0.0191/0.0145 and 3/2/130 decisions; all three ADAPT decisions are false adaptations. EATA has regrets 0.0024/0.0001/0.0342 and 108/1/26 decisions, including one false adaptation. These rows support candidate-level diagnosis, not a dataset-wide success label. The main paper discloses the nonstandard SAR operating point and the absence of an official-setting control.

B.4 Office-Home

Status: retrospective diagnostic; the five-checkpoint route is invalid.

B.4.1 Valid Descriptive Records

The primary Office-Home row (Venkateswara et al., 2017) is conservative. KGA regret is 0.0158, tying always-freeze (0.0158) and below always-adapt (0.0468), with 0 adapt decisions. This is a descriptive point-estimate match that evaluates conservative retention rather than selective routing. The archived protocol does not supply a predeclared uniform stability guarantee for the full-fit versus out-of-fold benefit predictor.

The 54-cell Office-Home stream-seed replication in Table 8 has KGA regret 0.0215 versus 0.0458 and 0.0217, a small point advantage relative to each fixed policy. The three stream seeds reuse the archived source checkpoint. The descriptive seed-bootstrap interval for improvement over always-freeze includes zero; the entire point gain comes from one ADAPT decision among 54 cells. This is not an interval-supported beats-both claim.

Table 10: Historical Office-Home diagnostics with equal weight per recorded condition; lower accuracy regret is better. The rows are separate experiments. The stream-seed replay uses a fixed SAR candidate. The five-checkpoint route is invalid, so only its post-hoc fixed-SAR opportunity regret (.0027) and freeze regret (.0230) remain; its actions have no routing or interval-support meaning. A/F/U denotes ADAPT/FREEZE/ABSTAIN. No interval-supported claim is made.

Protocol	n	Router / Adapt / Freeze	A/F/U
Archived stream-seed replay	54	.0215 / .0458 / .0217	1/14/39
Five-checkpoint candidate audit	15	invalid / .0027 / .0230	n/a

B.4.2 Invalid Five-Checkpoint Route

The post-hoc five-checkpoint Office-Home study is a candidate opportunity audit, not a valid KGA result. It uses five independently trained source checkpoints, the three target-test domains, one fixed IID/tiny condition, and Tent, EATA, and SAR. Checkpoint identity passed, but routing and serialization did not. The identifying agreement relation is binary-only and does not apply to the 65-class task; the old spectral estimate was sign-indeterminate and unbounded. Thirteen of 15 anchors were negative, 9 of 60 stored $\widehat{b}$ values fell outside $[0, 1]$, and EATA and SAR produced exactly the same predictions in 10 of 15 cells. The historical leave-one-out audit has only three cells per checkpoint, whereas a 90% empirical order-statistic single-candidate radius requires at least 10 total cells. All five result files fail strict JSON and contain 57 literal `Infinity` values in total.

The archived exploratory implementation printed ABSTAIN in all 15 cells, but those outputs have no safety, routing, or interval-support meaning. The valid descriptive quantities are checkpoint accuracies and candidate opportunity: freeze is 0.5711, SAR is 0.5914, their paired difference is 0.0203 (nominal post-hoc 95% t-interval $[0.0123, 0.0283]$), and the per-cell oracle is 0.5941. Oracle minus SAR is 0.0027 (nominal post-hoc 95% interval $[-0.0024, 0.0077]$). The best adapted candidate improves over freeze in 13 cells, ties in one, and harms in one. Because the target was already opened and SAR was chosen after comparing the audit means, these statistics are post-hoc opportunity evidence only.

B.5 Camelyon17

Status: opened, one-sided retrospective diagnostic.

The archived Camelyon17 OOD evaluation (Bandi et al., 2019) gives zero KGA and always-adapt regret, versus 0.1381 for freeze. All 18 cells are helpful. This opened panel is therefore a one-sided descriptive diagnostic, not prospective evidence of selective routing. The B–v2 within-seed diagnostics are separate. Their regenerated Tent/EATA/SAR KGA regrets are 0.0446/0.0260/0.0289, versus always-adapt regrets 0.0097/0.0040/0.0016 and always-freeze regrets 0.0820/0.0911/0.1001. None has lower regret than both fixed policies. The three run seeds do not identify independently trained checkpoints, and the target was already open. These rows remain stress diagnostics, not untouched hospital-domain evidence.

B.6 RxRx1

Status: retrospective retention diagnostic.

The displayed RxRx1 row (Sypetkowski et al., 2023) is model seed 0: KGA and freeze regret are zero against 0.2531 for always-adapt, with no adapt decisions. A separate audit across three independently trained model checkpoints repeats the same all-freeze tie. It strengthens model-variation reproducibility but adds no adapt exposure. RxRx1 therefore evaluates reproducible conservative retention rather than selective routing.

B.7 iWildCam

Status: invalid current metric contract; no canonical numerical row.

iWildCam (Koh et al., 2021) is not assigned a canonical numerical row. The archived scorer used sklearn macro-F1, which averages prediction-only classes, rather than the official WILDS metric, which averages labels present in the target truth. The archived runtime, sample identities, and target population were not sealed under the current contract, so its diagnostic scores and routing actions remain audit-only. A pinned rerun with the official metric, sealed sample identifiers, and a population hash is required before promotion. The older binary dominance flag and the later replay under the same invalid metric contract remain hash-locked only as superseded audit history.

B.8 Boundary and Transfer Diagnostics

Status: retrospective transfer diagnostics.

B.8.1 PACS

PACS (Li et al., 2017) KGA regret is 0.0431, compared with 0.0176 for always-adapt. The regret summary has 12 equal-weight domain–seed units. Its 216 action records are repeated decisions within those units, not 216 independent regret observations. The archive lacks the residual detail needed to replay the current gate, so PACS remains an aggregate performance diagnostic.

B.8.2 ImageNet-R Backbone Breakdown

ImageNet-R (Hendrycks et al., 2021) KGA regret is 0.0137, compared with 0.0064 for always-adapt and 0.0325 for always-freeze, across the ten-backbone architecture panel. No backbone has lower KGA point regret than each fixed comparator. KGA regret is higher than always-adapt regret on 8 of 10 backbones; four of those eight have a 0% harmful base rate, so routing has no harm to avoid. On `efficientnet_b0` (100% harmful) and `swin_t` (60.4% harmful), KGA improves over always-adapt but exactly matches always-freeze and makes no adapt decisions. The pattern is consistent with the regime map: routing can add value only where candidate harm is present, while these two rows evaluate conservative retention rather than selective routing.

B.8.3 CIFAR-10.1

CIFAR-10.1 (Recht et al., 2018) matches always-freeze at 0.0017 and is retained as a low-margin cross-seed transfer diagnostic. This pattern alone does not prove structural non-identifiability; weak evidence, low margin, estimator inadequacy, or calibration transfer can produce the same behavior.

B.9 Historical Protocol-Matched POEM and AETTA Ports

Status: historical local-port audit; not a synchronized official comparison. The repository retains one historical head-to-head artifact on the CIFAR-10-C Tent stream. Its KGA, POEM-style, and AETTA-style regrets are 0.00157361, 0.00880463, and 0.00732986; the associated always-adapt and always-freeze regrets are 0.00792338 and 0.12409792. Historical KGA-minus-port paired differences and unadjusted 95% paired-bootstrap intervals are

Port	Difference	95% interval
POEM-style	−0.00723102	[−0.00878038, −0.00570806]
AETTA-style	−0.00575625	[−0.00714260, −0.00439859]

Holm adjustment applies only to the archived p-values; the adjusted value for each of these two comparisons is 0.00039996.

Those estimates are protocol-matched diagnostics from local ports, not results from untouched official implementations. They also use an earlier KGA policy and are not synchronized with the current three-way cross-fit. The numerical ordering is therefore historical audit evidence only: it does not establish current-policy or official superiority. The source artifact is `experiments/kbound/results/mixed_headtohead_v1/HEADTOHEAD_RESULTS_cifar10c_tent_primary.json`; its stored `WIN` and `kga_beats` labels are withdrawn from the current claim set. A defensible new comparison would replay all methods on identical current-policy records and freeze the multiplicity family before scoring.

B.10 So2Sat Development and Action-Unit Audit

Status: development stop before target access.

The locked So2Sat-LCZ42 v1 study screened Tent and SAR in a development-only, multi-city, multi-checkpoint candidate-feasibility stage. It was not a target evaluation. Tent showed both helpful and harmful development cases, but neither candidate passed every predeclared eligibility check. The locked status is therefore *no feasible candidate; stop before gate calibration*. The detailed development authority and its binding receipt remain outside the default public build closure; the public release records only the protocol shape, stopping decision, and access boundary.

No held-out gate-calibration rows or target inputs, pixels, or labels were accessed. The study therefore contains no target natural-shift score. The gate-fit outcomes are open, so retuning on them cannot create fresh confirmatory evidence. V1 also declares one target action per city while its target runner creates city–checkpoint actions. This mismatch did not affect the stop because no target action was created. Prospective v2 resolves the action unit as city–checkpoint and supplies a source-complete runner. It remains a no-go: no v2 pre-calibration seal or target authorization exists, and gate calibration and target evaluation remain unopened. V2 does not retroactively change the negative v1 record.

B.11 DomainNet Painting Development and Fixed Diagnostics

These additional results are descriptive on already-opened painting development. The five-checkpoint cohort uses 24 shared check cells per checkpoint: its 120 seed-cell records are not 120 independent environments. A separate reference-source ResNet-50 checkpoint is used by two different recipes. The local entropy probe updates BatchNorm affine parameters using one unlabeled pass, Adam at 10^{-3}, and batches of at most 32; it restores source BatchNorm running buffers before evaluation. It is neither online Tent nor AdaContrast. The AdaContrast-derived single-process bridge uses batch size 16, 15 epochs, square-resize preprocessing and direct-logit inference; it is not an exact reproduction of the published pipeline. Each reference-source check panel contains 24 cells and 3,073 image evaluations. Their frozen/candidate correct totals are 1387/1366 for R50 entropy and 1384/1131 for the bridge. The differing frozen totals mean this is not a matched head-to-head comparison.

Table 11: Opened DomainNet painting development, kept separate from the canonical primary panel. Regret and radii are in percentage points. Repeated seed/cell records and the fixed diagnostic cells do not provide independent prospective trials.

Saved evaluation	Units and measured effects	Gate or diagnostic result
Five-checkpoint recipe	120 seed-cell records; 12 helpful, 15 harmful, 93 tied	120 ABSTAIN; equal-seed regret 0.1106770833 for gate/freeze, 0.1235465116 for adapt
R50 local entropy	24 check cells; 7 helpful, 14 harmful, 3 tied	24 ABSTAIN; radius 2.6099386731; inclusion 21/24 (87.5%); regret 0.390625 gate/freeze, 1.07421875 adapt
AdaContrast-derived bridge	24 check cells; 1 helpful, 23 harmful, 0 tied	24 ABSTAIN; radius 10.8115435382; inclusion 24/24; regret 0.0651041667 gate/freeze, 8.2995195413 adapt
Fixed queue diagnostic	First 4 DEV_fit cells; 512 paired image evaluations	Correct totals: capacity 16384, 187; capacity 64, 181; freeze, 243. Smaller queue loses 6/512 (1.171875 points); 3 cells worse, 1 better
Fixed BN decomposition	Same 4 DEV_fit cells and 512 evaluations	Correct totals: freeze 243; stats-only 107; other-state-only 133; actual adaptation 187. Every altered condition harms all 4 cells

All three check evaluations have zero ADAPT and zero FREEZE exposure. Their conditional false-ADAPT and false-FREEZE rates are undefined; zero false commitments under universal abstention supply no directional exposure. The inclusion rates are descriptive, not validation of nominal coverage on unseen groups. These development results do not establish a selective-routing gain over freeze.

The queue intervention changed capacity, negative-pool size, memory age, and the inherited class-aware masking interaction; it does not isolate random initialization. The BN experiment crossed original/adapted running buffers with original/adapted remaining state after identical baseline training; other state includes BN affine parameters. Its accuracy-scale interaction is +37.109375 percentage points. These non-additive effects do not identify a BN-only cause of harm; restoring source statistics alone does not rescue the adapted model. Neither follow-up selects a new adapter, radius, or decision rule. Any retrospective radius or predictor checks remain exploratory.

B.12 Office-Home mechanism check

Office-Home: archival selection and changed-recipe smoke. On the opened 54-cell stream-seed archive, the post-hoc rule "serve SAR iff its recorded affine update norm equals zero exactly" selects 35 candidates and freezes 19 cells. Its accuracy regret is 0.00035613, versus 0.04579772 for always-adapt and 0.02172365 for always-freeze. This rule is not the calibrated KGA gate, and choosing it after inspecting the archive does not constitute a new held-out result. A zero affine norm also does not imply frozen predictions: candidate inference can replace source BatchNorm statistics with current-batch statistics.

A separate MPS execution smoke used one class-balanced validation episode in each of Art, Clipart and Product: 65 evaluation images and 16 disjoint adaptation images per domain. All three arms used the same 65-image evaluation batch. They were frozen source statistics, batch statistics without parameter updates, and the local SAR helper with one pass over two batches of eight, learning rate 10^{-3}, $\rho = 0.05$, entropy threshold $0.4 \log(65)$ and reset threshold 0.2. This deliberately smaller recipe differs from the archived aggressive SAR recipe. The checkpoint is identified by observed local bytes and metadata; its published training recipe was not authenticated.

Table 12: Three-arm Office-Home execution smoke: correct predictions out of 65 per domain. These are three development episodes, not independent confirmatory trials.

Domain	Frozen	Batch stats	SAR	SAR steps
Art	37	34	34	0
Clipart	26	30	30	0
Product	41	43	43	1

Batch-statistics-only and SAR predictions agree on all 195 evaluation images, including Product, where one optimizer step was accepted and the affine update norm was nonzero. Their totals are both 107/195, versus 104/195 for frozen inference. The archival exact-zero rule applied descriptively to this smoke yields 105/195, below always-SAR. The smoke therefore does not reproduce the archival selector's advantage. Source tensors and buffers were preserved, and all prediction receipts were written before scoring. No KGA predictor or radius was fitted, calibrated or evaluated. Passing the smoke establishes execution of the three arms and one SAR update branch; it does not establish statistical efficacy or the cause of historical gains.

The runner-recorded interval was 20.43 seconds, beginning after argument parsing and output-directory creation and excluding process startup and module imports. It includes subsequent setup and integrity work as well as execution of all three arms. Per-arm latency, tail latency, sustained throughput and peak memory were not measured; the saved before/after allocator readings are snapshots, not a peak-memory estimate.

B.13 Withdrawn Development Proxy for the Population Budget

Status: negative retrospective diagnostic; estimation route withdrawn.

Without additional structural information, label-free evidence cannot uniformly validate a residual bound below the unresolved radius of its compatible fibre. This does not exclude a valid constant bound at or above that radius, and $\beta = 0$ is not a safe default. The tested source-like development proxy was 1.4–50× too small on real evaluation cells: 24–73% fell outside $\mathcal{C}_\beta$, commit error was 0.4–16.6%, and five of ten configurations returned zero commitments on all 405 ImageNet-C cells. The proxy did not observe the theorem's disagreement-conditional residual, and its named raw artifact is not regenerable from the canonical bundle. We therefore withdraw estimation and require domain knowledge, a declared coupling, or historical labeled deployments under a transfer assumption.

The population layer targets Δ, whereas empirical calibration targets Δ^{cell}. Appendix A states the additional sampling argument needed to bridge them. Neither layer establishes risk alignment, estimator transfer, or residual coverage in a new domain.

C Compact Assumption and Claim Ledger

Table 13 separates formal claims from empirical findings and records the conditions and scope limits for each.

Table 13: Formal half of the claim-to-support ledger. Each row separates its target and assumptions from what is and is not supported.

Statement and target	Required assumption or protocol	Supported conclusion	Not supported
Disagreement reduction; population Δ	binary labels, 0/1 loss, fixed pair, $\mu_T(D) > 0$	$\Delta = 2\mu_T(D)(M+\gamma)$	calibration transfer or observability of γ
Interior ambiguity; population Δ	Thm. 1; $\beta > 0$, $\lvert M\rvert < \beta$, fibre richness	evidence-identical laws with opposite nonzero benefit	ambiguity outside the declared class
Closed band; population Δ	Prop. 1; strict semantics and richness	no uniformly sound strict action on $\lvert M\rvert \leq \beta$	unequal task risk at every boundary law
Strict frontier; population Δ	Thm. 2; fixed evidence and declared $\mathcal{C}_\beta$	pointwise maximal sound three-action rule	learning β from the unlabeled batch
Audit floor; residual radius	nonempty fibre, bounded attainable residuals, measurable audit, common joint evidence/randomization law, and uniform validity	$\widehat{\beta} \geq \Gamma_z$ with the stated probability	a finite-batch estimator or data-learned β
Coverage-to-action; declared cell target B	Prop. 2 and Eq. (9)	marginal false ADAPT and false FREEZE control for B	population-risk, conditional, group, or anytime control

Table 14: Empirical and future-evidence half of the claim-to-support ledger. Evaluation units and statuses remain protocol-specific; no row upgrades retrospective evidence.

Statement and target	Required assumption or protocol	Supported conclusion	Not supported
CIFAR-10-C; measured cells	opened one-checkpoint grid and outcome-disjoint cross-fit	candidate-specific actions, errors, and point regret; Tent nominal intervals favorable	prospective confirmation; six-contrast Holm rejection
CCT-20; measured cells	locked 45-cell protocol and declared utility endpoint	all-frozen utility preservation with 0/44/1 actions	selective routing, improvement over always-freeze, or population safety
Other natural replays; measured cells	Tables 8–9; fixed splits and candidates	protocol-specific retention or adverse behavior	one pooled success claim
So2Sat; development eligibility	Sec. 8.4; locked feasibility rules	no candidate passed the pre-target screen	gate calibration or a target score
DomainNet; opened development	Sec. B.11; shared cells, separate candidates and fixed diagnostics	universal abstention and adverse diagnostic outcomes	independent confirmation, pooled-trial inference, or selective-routing success
ImageNet-C; measured cells	Table 8; candidates and operating points kept separate	candidate-specific diagnostic evidence	dataset-wide or published-default conclusion
Other replay scope; recorded units	Tables 8–9	observed protocol behavior	structural unknowability or universal transfer
Prospective confirmation; future targets	Sec. K; diverse unopened environments	a frozen evaluation design	a current empirical result
Physical camera; future sessions	locked session split and held-out recordings	protocol infrastructure only	a completed camera result

D Calibration and Evaluation Contracts

D.1 Controlled-Grid Contract

The CIFAR-10-C grid is an opened, dependent, constructed six-family diagnostic. Its fitting unit is the condition cell. Stable identifiers assign deterministic score folds. For each fold, its complement is divided into disjoint estimator-fit and residual-calibration subsets. The fitted predictor and exact-rank radius are then applied only to the score fold. Consequently, the scored outcome $B_i = \Delta_i^{\text{cell}}$ enters neither the fit nor the calibration set that determines its own prediction, radius, or action. This is a three-way cell-outcome-disjoint cross-fit, not leave-one-cell-out calibration.

The CIFAR-10-C grid uses one ResNet-18 checkpoint and five RNG stream/run seeds. Within each seed it crosses six corruption families, severities 1 and 5, batch sizes 200/16/8, IID/imbalanced/single-class composition, two update schedules (10 steps at 10^{-3} and 50 steps at 2.5×10^{-3}), and two repeats: 432 cells per seed and 2,160 per candidate. Point summaries weight cells equally. The family analysis first averages seeds and cells within a family and then weights the six family means equally. These are opened, dependent, constructed cells coupled through one corpus and checkpoint; the cross-fit does not make them exchangeable or validate coverage on unseen environments.

Tent, EATA, and SAR are local protocol-matched ports. Candidate preview runs in training mode with BatchNorm running statistics disabled, so the measured delta combines evaluation-batch statistic replacement with gradient updates to normalization affine parameters. The two recorded update schedules are part of the grid. They are not untouched reproductions of each method's published default operating point; the main text gives the additional SAR-specific departure.

D.2 Natural-Shift Contract

Development conditions fit the benefit model; a disjoint calibration partition supplies residuals; choices are frozen; and the target partition is scored once where the archived protocol records all three stages. When the saved evidence does not support that complete chain, the row is marked descriptive or diagnostic. Replaying a table does not strengthen its original design.

The candidate TTA step itself is transductive. Episodic candidates update on each unlabeled evaluation batch before predicting that batch. Online candidates update on a separate auxiliary stream, but prediction still uses evaluation-batch BatchNorm statistics. Thus "disjoint" above refers to development, residual-calibration, and target partitions (and to the recorded auxiliary stream versus evaluation sample identities); it does not mean that candidate inference is inductive. No target labels enter adaptation, prediction, evidence, or the action.

D.3 Action Contract

Every action is computed from Z, $\widehat{\Delta}$, and ε without target labels. Offline labels determine Δ^{cell}, the measured oracle action, empirical regret, and false-commit indicators only after the action is fixed. Existing artifact fields named `delta` store this cell outcome; their schema and numerical values are unchanged. Abstention resolves to the frozen prediction for measured task-performance accounting, while its log retains the distinction from an interval-supported FREEZE decision.

The whole-pipeline invariant is tested by perturbing one scored B_i and rerunning all fits. Its $\widehat{\Delta}_i$, ε_i, and action must remain unchanged. The maintained test `tests/test_controlled_grid_crossfit.py` checks this for the implementation and also exhibits the old radius leak as a negative control. This is a software non-leakage result for the scored cell; it does not prove exchangeability, estimator validity, or future-domain coverage.

D.4 Configuration Contract

A configuration identity binds the checkpoint, adapter and update settings, evidence schema, benefit estimator, residual construction, α, and action unit. Section H records that identity once. Any mismatch requires a declared transfer argument or recalibration; otherwise the controller does not commit.

E Decision Exposure and Repetition Audit

Point regret alone can conceal whether a conservative match exercised the ADAPT branch. Table 15 therefore reconstructs action counts from the same canonical panel used in the main-paper primary CIFAR table 5 and Tables 8 and 9. The three additional DomainNet rows use the separate saved development summaries in Sec. B.11; they do not change that canonical panel or form a pooled set of independent trials. A/F/U denote adapt, freeze, and abstain. PACS is omitted from the action table because the released aggregate cross-checks regret but does not contain the per-cell benefit estimates and residuals required for a complete gate replay. We omit iWildCam because its archived actions were scored under the wrong macro-F1 contract.

Table 15: Action exposure in replayable rows. A/F/U denotes ADAPT/FREEZE/ABSTAIN. Counts are unweighted evaluation cells, not independent experiments; regret tables use the weighting declared by each protocol. False A/A and false F/F use the corresponding action denominator. A dash means that the direction was unexposed or was not reconstructed by the maintained authority; it does not mean zero error. Controlled-grid rows use the current three-way cross-fit; the remaining rows retain their stated diagnostic protocol. The canonical numerical panel is bound to SHA-256 `35d4c165843de1ece3cb35ffb4ac50dbfcbfa33b646754e2b6d133fbdfa78c6e`; its input inventory is bound separately by the source manifest. DomainNet rows have separate development authorities and are not part of that canonical panel.

Row	n	A	F	U	false A/A	false F/F	repetition / scope
Office–Home primary	35	0	11	24	–	–	2 test-stream seeds; ties freeze
Office–Home stream-seed repl.	54	1	14	39	0/1	–	3 test-stream seeds; point-only edge
ImageNet–C SAR	135	28	9	98	0/28	–	5 seeds × 27 cells
ImageNet–C Tent	135	3	2	130	3/3	–	5 seeds × 27 cells
ImageNet–C EATA	135	108	1	26	1/108	–	5 seeds × 27 cells
ImageNet–R	480	179	34	267	6/179	–	4 seeds × 10 backbones × 12 cells
CIFAR–10–C Tent	2160	1091	337	732	2/1091	1/337	5 seeds × 432 cells
CIFAR–10–C EATA	2160	1207	105	848	0/1207	0/105	5 seeds × 432 cells
CIFAR–10–C SAR	2160	1414	0	746	0/1414	–	no FREEZE exposure
CCT–20	45	0	44	1	–	1/44	5 checkpoints × 9 locations
DomainNet five-checkpoint	120	0	0	120	–	–	5 checkpoints × 24 shared development cells
DomainNet R50 entropy	24	0	0	24	–	–	separate opened development evaluation
DomainNet AdaContrast-derived	24	0	0	24	–	–	separate opened development evaluation
Camelyon17 OOD	18	18	0	0	0/18	–	opened helpful-only OOD subset
Camelyon17 B–v2 Tent	108	27	0	81	0/27	–	3 within-seed grids; diagnostic
Camelyon17 B–v2 EATA	108	51	0	57	3/51	–	3 within-seed grids; diagnostic
Camelyon17 B–v2 SAR	108	52	0	56	1/52	–	3 within-seed grids; diagnostic
RxRx1 J	60	0	60	0	–	–	model seed 0; 3-checkpoint audit also all-freeze
CIFAR–10.1 K	48	0	22	26	–	–	locked cross-seed replay

The largest adapt exposure occurs on CIFAR-10-C, making it the main diagnostic of the adapt branch. Tent records two false adaptations among 1,091 ADAPT decisions and one false FREEZE among 337 FREEZE decisions; EATA and SAR record no false adaptations, although SAR never exposes FREEZE. The CCT-20 row supplies the complementary all-frozen case: one false FREEZE among 44 FREEZE decisions and no ADAPT exposure. Primary Office-Home and RxRx1 have zero adapt exposure, while the 54-cell Office-Home replication has one adapt decision. Those rows chiefly test conservative retention. The invalid five-checkpoint Office-Home route and the invalid-metric iWildCam archive are excluded. Camelyon17 OOD has the opposite limitation: every archived cell is helpful and every action adapts. On ImageNet-C, Tent makes three ADAPT decisions and all are false adaptations, whereas EATA makes 108 ADAPT decisions with one false adaptation and SAR makes 28 with none. Similar regret values can therefore hide different operating behavior.

The repetition column also prevents a common inflation of evidence. Five seeds of one checkpoint and grid protocol are not five independently designed studies. Ten ImageNet-R backbones are a panel of candidate policies, not ten target domains for one frozen gate. Three Camelyon17 within-seed grids permit a stability diagnostic but do not create an untouched hospital-domain replication. RxRx1 is stronger on model variation because the source checkpoint is independently trained three times, but its all-freeze behavior still provides no adapt exposure.

F Inference Discipline

F.1 Point Estimates, Paired Gaps, and Clusters

For a baseline $b \in \{\text{always-adapt}, \text{always-freeze}\}$, the paired routing gap at condition j is

$$G_{bj} = \text{regret}_{b,j} - \text{regret}_{\text{KGA},j}\,.$$

A positive mean gap means lower regret for KGA. All paired routing-gap intervals reported in this paper use this baseline-minus-KGA sign convention. The point-estimate two-comparator criterion requires both means to be positive. Interval support additionally requires the lower confidence limit for both paired means to exceed zero at the declared resampling unit. These definitions are intentionally separate: Table 8 can establish a point ordering, but not an inference claim without the associated unit, interval artifact, and justified coverage assumptions. A positive computed lower endpoint is numerical interval support under the resampling model, not verified coverage. Multiplicity adjustments do not repair invalid marginal intervals or tests.

Condition-level resampling treats cells as exchangeable and can understate uncertainty when severities and compositions share a corruption family. Corruption-family resampling preserves observed within-family dependence, but few clusters limit reliability; conservativeness is not guaranteed. Seed resampling measures variation across repeated training or test streams only when the seeds represent the intended deployment randomness. The compact paper therefore does not use one generic “bootstrap CI” label. An inference claim must state whether the unit is cell, corruption–severity pair, corruption family, seed, backbone, domain, or physical session.

For a vector of cluster gaps $G = (G_1, \ldots, G_m)$, the exact sign-flip reference calculation uses the joint null condition

$$G \overset{d}{=} s \odot G \quad \text{for every } s \in \{-1, +1\}^m. \tag{16}$$

Independent zero-symmetric cluster gaps are sufficient. Exchangeability, marginal symmetry, or one global sign symmetry alone is not sufficient. Enumeration removes Monte Carlo error but does not establish this null condition.

F.2 Current-Policy Corruption-Family Sensitivity

The retrospective family contains all six candidate-by-baseline contrasts. The replay policy, corruption-family unit, sign-flip reference test, and Holm adjustment were selected after the underlying CIFAR outcomes were open. The current-policy artifact averages run seeds and condition cells within each of the six families. Tent’s nominal intervals favor KGA against always-adapt ($[0.002087, 0.009537]$) and always-freeze ($[0.062822, 0.177573]$). EATA’s adapt-side interval crosses zero ($[-0.001348, 0.004382]$), although its freeze-side interval is positive. SAR is worse than always-adapt. These nominal bootstrap intervals are distinct from the retrospective sign-flip/Holm analysis. The within-Tent two-contrast Holm calculation is itself post hoc and is not the six-test family reported in Table 16.

Table 16: Retrospective sensitivity over six opened CIFAR-10-C corruption families. Each candidate uses 2,160 cells from one checkpoint; seeds and conditions are averaged within family, and the six family means receive equal weight. Entries are baseline regret minus KGA regret with nominal, unadjusted 95% family-bootstrap intervals; positive values favor KGA. Holm values adjust the six retrospective sign-flip reference tests. For Tent, the post-hoc within-candidate Holm values are 0.046875 (adapt) and 0.03125 (freeze), while the six-contrast values are 0.140625 and 0.09375. The replay, unit, tests, and adjustment were not preregistered. The table supplies no verified coverage, independent-checkpoint inference, prospective confirmation, natural-shift evidence, or official POEM/AETTA comparison.

Candidate	Adapt gap [95% CI]	Freeze gap [95% CI]	Six-way Holm p (A/F)
Tent	0.00606 [0.00209, 0.00954]	0.12202 [0.06282, 0.17757]	0.14062/0.09375
EATA	0.00164 [-0.00135, 0.00438]	0.12970 [0.06893, 0.18746]	0.31250/0.09375
SAR	-0.00145 [-0.00294, -0.00027]	0.13869 [0.07972, 0.19683]	0.90625/0.09375

F.3 Retrospective Interval Inclusion by Family

This audit summarizes the stored per-cell authority produced by the three-way cross-fit. For every scored cell, its measured outcome was absent from both estimator fitting and residual calibration. The audit performs no tuning, new target access, or training; it verifies the stored predictions, radii, actions, and input/code/output digests.

The observed inclusion counts are 1,976/2,160 for Tent, 1,950/2,160 for EATA, and 1,949/2,160 for SAR. Tent also has one false FREEZE among 337 FREEZE decisions. These are in-sample cross-fit diagnostics on opened, dependent cells, not independent estimates for new environments. Family rows localize misses and widths but do not supply group-conditional guarantees. The nominal 90% setting is therefore not a validated coverage promise.

Table 17: Overall retrospective interval diagnostics for 2,160 equally weighted cells per CIFAR-10-C candidate. Inclusion concerns measured cell benefit, not population benefit. Marginal directional errors divide by all cells; conditional errors divide by the corresponding action count and are undefined when that action is absent.

Candidate	n	Nominal	Inclusion	Mean width	Median width	Commitment	FA_u/FA_c	FF_u/FF_c
TENT	2160	0.9000	0.9148	0.0478	0.0457	0.6611	0.0009/0.0018	0.0005/0.0030
EATA	2160	0.9000	0.9028	0.0397	0.0389	0.6074	0.0000/0.0000	0.0000/0.0000
SAR	2160	0.9000	0.9023	0.0286	0.0290	0.6546	0.0000/0.0000	0.0000/–

Table 18: Current-policy interval and decision diagnostics by corruption family. Each row aggregates equal-weight opened cells after the three-way cross-fit; the scored cell outcome enters neither its fit nor calibration set. Width is 2ε, and A/F/U means ADAPT/FREEZE/ABSTAIN. Conditional errors use the corresponding action count as denominator and are undefined when that action is absent. The table is retrospective and is not held-out group-coverage validation.

Candidate	Family	n	Inclusion	Mean width	Commitment	False A/A; FA_c	False F/F; FF_c
TENT	contrast	360	0.8361	0.0478	0.6917	0/180; 0.0000	0/69; 0.0000
TENT	defocus_blur	360	0.9417	0.0480	0.7083	0/180; 0.0000	0/75; 0.0000
TENT	fog	360	0.9472	0.0480	0.6917	0/180; 0.0000	0/69; 0.0000
TENT	gaussian_noise	360	0.8667	0.0480	0.8556	0/305; 0.0000	1/3; 0.3333
TENT	jpeg_compression	360	0.9694	0.0478	0.3472	2/66; 0.0303	0/59; 0.0000
TENT	pixelate	360	0.9278	0.0475	0.6722	0/180; 0.0000	0/62; 0.0000
EATA	contrast	360	0.8417	0.0396	0.5694	0/180; 0.0000	0/25; 0.0000
EATA	defocus_blur	360	0.9278	0.0398	0.5694	0/180; 0.0000	0/25; 0.0000
EATA	fog	360	0.9139	0.0399	0.5750	0/180; 0.0000	0/27; 0.0000
EATA	gaussian_noise	360	0.8528	0.0399	0.9722	0/350; 0.0000	0/0; –
EATA	jpeg_compression	360	0.9694	0.0396	0.4056	0/137; 0.0000	0/9; 0.0000
EATA	pixelate	360	0.9111	0.0393	0.5528	0/180; 0.0000	0/19; 0.0000
SAR	contrast	360	0.8694	0.0287	0.5000	0/180; 0.0000	0/0; –
SAR	defocus_blur	360	0.9306	0.0287	0.5000	0/180; 0.0000	0/0; –
SAR	fog	360	0.8944	0.0283	0.5000	0/180; 0.0000	0/0; –
SAR	gaussian_noise	360	0.8417	0.0286	1.0000	0/360; 0.0000	0/0; –
SAR	jpeg_compression	360	0.9667	0.0283	0.8944	0/322; 0.0000	0/0; –
SAR	pixelate	360	0.9111	0.0288	0.5333	0/192; 0.0000	0/0; –

F.4 False-Adapt and False-Freeze Accounting

The event in Eq. (4) is unconditional and concerns measured cell benefit. Its reported frequency divides ADAPT decisions with $\Delta_j^{\mathrm{cell}} \leq 0$ by all evaluated units, including freezes and abstentions. This need not estimate the population-harm event $\{g = \textsc{adapt}, \Delta \leq 0\}$. Conditional false-adapt divides by adapt decisions and is undefined when no adaptation occurs. A zero observed count should be accompanied by the number of adapt decisions; otherwise a controller that never adapts appears perfect. Table 15 exposes that denominator.

False FREEZE is the symmetric event: the policy chooses FREEZE when measured cell benefit is nonnegative. Its marginal rate divides by all evaluated units, whereas its conditional rate divides by FREEZE exposure. The latter is undefined when the policy never freezes, as in the current SAR grid. Every directional-error statement therefore appears with its action denominator.

The three-way split removes the direct path from B_i to its own prediction, radius, and action. It does not make the overlapping fitted models or the constructed cells independent. Pooled cross-fit inclusion therefore remains a descriptive reuse of the opened panel and supplies neither exchangeability nor a finite-sample guarantee for the next corruption family. A genuinely held-out family or environment remains a different protocol.

F.5 Multiplicity and Search History

The repository contains many exploratory configurations. One superseded census reported 1,387 determinations and 326 positives (23.5%); a broader archived scan reported 1,427 determinations and 345 positives (24.2%). Neither census froze one family definition, file list, and commit, so neither is the release multiplicity family. The reconciled panel is also post hoc, although some component protocols were individually locked. Confirmatory work must declare every candidate–track test, calibration and target unit, seed structure, inference unit, and two-comparator criterion before labels are opened, while retaining failures and incomplete runs.

G Track-Level Provenance and CCT Ledger

Separate archival records preserve two additional scope checks outside the canonical panel. FMoW Protocol L reached held-out scoring and matched always-freeze, while always-adapt had lower regret; its status remains not-cleared. PovertyMap stopped at its preregistered development screen because the harm-detection AUC was 0.637, below the 0.65 threshold, so held-out validation and test scoring did not run. These records remain in the claim inventory and are not pooled with the canonical rows.

The provenance table is part of the result, not administrative metadata. A reviewer can trace a canonical row to compact records and their original archive hashes. If the original large artifact is not shipped, the manifest still records its byte count and digest. This guards against silent replacement of a result while allowing the release to omit raw datasets and oversized transient checkpoints.

Table 19: Release-status inventory, not a performance comparison. "Replayable" means stored quantities suffice to recompute the displayed gate output under the named scorer; it does not imply fresh data, independent units, or confirmatory status. Units and weighting remain protocol-specific.

Track	Released status	Present evidence	Remaining limitation
CIFAR-10-C	replayable diagnostic	per-cell Δ^{cell}, $\widehat{\Delta}$, radius, action; 5 RNG seeds; 3 candidates	opened, dependent, constructed grid from one checkpoint; no untouched family
ImageNet-C	replayable diagnostic	per-condition records for 5 seeds and Tent/EATA/SAR	opened 27-cell controlled configuration; candidate-specific, not one policy
Office-Home	replayable	canonical primary and 54-cell stream-seed records; later five-checkpoint audit reported separately	full-fit-versus-OOF transfer-stability premise not predeclared
iWildCam	invalid archived diagnostic	opened test-split records scored with the wrong macro-F1 definition	no completed current-protocol or official-metric result; population and overlap were not sealed
Camelyon17	replayable diagnostic	archived opened OOD row plus separate B–v2 diagnostics	OOD row is one-sided and non-prospective; B–v2 is not untouched-domain evidence
RxRx1	replayable	displayed model-seed-0 row plus three model-checkpoint records and source hashes	all three checkpoints freeze throughout; no helpful test conditions
CCT-20	release-complete; cell-outcome-unopened, internally sealed execution	5 independent checkpoints, 9 target locations, actions, predictions, score, manifest, and receipts	aggregate target-label metadata had previously been inspected; not globally label-unopened or publicly preregistered; only two fit locations; no ADAPT exposure; trans-target extrapolation remains an assumption
So2Sat-LCZ42	sealed development stop	Tent and SAR candidate bundles; multi-city, multi-checkpoint gate-fit screen; no gate-calibration or target access	no feasible candidate; gate-fit outcomes are open; no gate calibration or target result
PACS	partial	three-seed aggregate and source cross-checks	missing per-cell $\widehat{\Delta}$ and residuals prevent full gate replay
ImageNet-R	replayable	4 seeds, 10 candidate backbones, 12 conditions each	architecture aggregate is diagnostic, not one deployable router
CIFAR-10.1	replayable	locked cross-seed condition records	low-margin conservative match; selective routing not exercised
Physical camera	protocol only	dashboard, logging schema, and planned session split	no completed fresh recording, held-out session, or result table

G.1 CCT-20 Inference, Location, and Release Ledger

Provenance ledger. This was a cell-outcome-unopened, internally sealed execution: individual cell outcomes remained unopened until one-shot scoring. Aggregate target-label metadata had previously been inspected, so the study was not globally label-unopened or publicly preregistered. Item-level target labels were deserialized once, after the one-shot scoring marker was spent. The release chain also records the numeric-v2 repair in a sealed addendum created before outcome access. That repair replaced an

Table 20: Locked CCT-20 primary contrasts over 45 equal-weight checkpoint–location cells from five independent checkpoints and nine location clusters. The estimand is baseline regret minus KGA regret, so positive values favor KGA. A paired two-way product bootstrap resamples checkpoint rows and location columns. Two nominal 97.5% intervals target a nominal 95% Bonferroni family level; actual family coverage still requires valid marginals. Reference p-values enumerate one-sided sign flips of nine location means, with Holm adjustment. This cell-outcome-unopened execution reproduces the locked calculation but does not establish a population guarantee.

Comparator	Baseline regret − KGA regret	Nominal 97.5% CI	Sign-flip p	Holm p	Holm flag (.05)
Always adapt	0.18190227	[0.08501071, 0.31066374]	0.00195312	0.00390625	yes
Always freeze	0	[0, 0]	1	1	no

unstable divergence calculation with a stable Gaussian KL computation; it did not change the model, data split, or scoring rule. We treat it as protocol provenance, not result-driven tuning. All primary outcomes and action/effect counts come from `paper/generated/cct20_release_manifest.json`. Its sidecar `paper/generated/cct20_release_manifest.json.receipt.json` binds the file-level digest; the generated macros instead record canonicalized inference-document SHA-256 `5d1ba74d9a8b16ef24746ca3d95969e635d34fd4112806f6bf91b4b45a6cf907`.

Locked thresholds. Strong success required all six conditions: all 45 cells complete; both nominal 97.5% Bonferroni intervals strictly above zero; both exact nine-location sign-flip tests surviving Holm at 0.05; ADAPT rate at least 0.10; FREEZE rate at least 0.10; and total commitment at least 0.30. Completion, freeze exposure, and commitment passed. The interval and test requirements failed against always-freeze, and ADAPT exposure was zero. The release therefore fails the strong-success criterion.

The nominal pointwise 95% intervals in the CCT utility-preservation table 6 belong to the locked utility-preservation endpoint. The nominal 97.5% Bonferroni intervals in the strong-success audit in Table 20 belong to a separate, stricter two-comparison criterion; they are not alternate versions of the same interval.

Finite-sample sign-flip validity requires the null joint law of the nine location gaps to satisfy Eq. (16). Independent zero-symmetric location gaps conditional on the evaluated checkpoints would suffice, but that assumption is unverified here. Exchangeability alone does not suffice. The sign-flip analysis conditions on the evaluated checkpoints, whereas the two-way bootstrap also resamples checkpoint levels. These are different uncertainty models, and neither establishes distribution-free safety for unseen locations.

Primary result. KGA exactly matched always-freeze. Its contrast against always-freeze was 0 with nominal Bonferroni-adjusted 97.5% bootstrap interval [0, 0], enumerated location sign-flip $p = 1$, and Holm $p = 1$. Against always-adapt, the contrast was 0.18190227 with interval [0.08501071, 0.31066374], sign-flip reference $p = 0.00195312$, and Holm $p = 0.00390625$. Measured cell benefit is adapted accuracy minus frozen accuracy: 1 cell was helpful, 0 were exactly zero, and 44 were harmful. These counts explain why freezing was useful here. They do not show that the gate learned when to adapt. KGA made 0/44/1 ADAPT/FREEZE/ABSTAIN decisions. All 45 served predictions used the frozen model, but the sole helpful cell received an interval-supported FREEZE, yielding one false FREEZE among 44 FREEZE actions (1/45 overall). This is all-frozen fail-closed service. It passes only the locked utility-preservation rule and does not demonstrate selective routing or strong success.

Limits and next step. All 55 development cells showed lower performance after Tent than under the frozen model, so the gate had no helpful ADAPT example. The ridge fit used 10 checkpoint–location

Table 21: Complete cell-outcome-unopened CCT-20 location ledger. A/F/U denotes ADAPT/FREEZE/ABSTAIN, and +/0/− counts positive/zero/negative measured-benefit cells. Difference columns are equal-weight means over five checkpoint cells; positive policy differences favor KGA. Values are rounded to four decimals, while the sealed manifest retains full precision. Cells are repeated-model observations, and the inferential cluster unit is the camera location.

Location	n/checkpoint	A/F/U	Benefit +/0/−	Adapt − freeze	KGA − adapt	KGA − freeze
0	1302	0/5/0	0/0/5	-0.5418	0.5418	0.0000
7	1283	0/5/0	1/0/4	-0.0293	0.0293	0.0000
28	911	0/5/0	0/0/5	-0.0562	0.0562	0.0000
40	800	0/5/0	0/0/5	-0.1232	0.1232	0.0000
46	4432	0/4/1	0/0/5	-0.0723	0.0723	0.0000
78	1633	0/5/0	0/0/5	-0.0839	0.0839	0.0000
100	3441	0/5/0	0/0/5	-0.2035	0.2035	0.0000
105	1540	0/5/0	0/0/5	-0.2892	0.2892	0.0000
130	983	0/5/0	0/0/5	-0.2376	0.2376	0.0000

rows from only two camera units for 11 recorded features. Calibration used 45 rows collapsed to nine calibration camera locations, while the locked targets came from held-out target locations. Coverage therefore needs an unverified cross-split exchangeability assumption. Target Mahalanobis diagnostics ranged from 7.12 to 485.67 (median 28.73), showing strong extrapolation; this diagnostic never changed an action and is not a formal coverage guarantee. One natural target study also cannot establish transfer of selective routing across natural shifts. Cell-level 16-indicator macro-F1 is archived as descriptive secondary evidence; no post-hoc aggregate or inference claim is made.

Table 21 reports all nine location clusters. Evaluation n is the number of evaluation images per checkpoint at that location. Each A/F/U triplet and each helpful/zero/harmful triplet therefore has denominator five, one cell for each checkpoint. These labels describe measured cell outcomes. Accuracy differences use one sign throughout: adapted minus frozen for adaptation benefit, and KGA minus the named fixed policy for the two policy columns.

The release ledger contains all 45 cells: 0 ADAPT, 44 FREEZE, and 1 ABSTAIN actions; 1 helpful, 0 zero, and 44 harmful adaptation effects. The release verdict remains SAFE UTILITY ONLY. The location table is descriptive; sign-flip enumeration uses the nine location means under the reference model in Eq. (16), and no post-hoc aggregate is claimed for the secondary macro-F1 outcome.

H Extended KGA Implementation Contract

H.1 State Isolation

The frozen and candidate predictors must not share mutable normalization buffers, optimizer state, or parameter storage during preview. The candidate is created from a checkpoint, adapted on the unlabeled window, and evaluated without overwriting the active state. Only an adapt decision may replace the active checkpoint. Freeze and abstain discard the temporary state. A proposed deployment test should hash the active parameters before and after each non-adapt decision and require equality.

H.2 Evidence Determinism

Evidence extraction must define preprocessing, batch order, numerical precision, class ordering, normalization constants, and missing-value behavior. Paired features require frozen and candidate outputs on the same samples in the same order. Source statistics and ATC thresholds are immutable calibration artifacts. Evidence schema versions should be explicit rather than inferred from vector length, because two schemas can share a dimension while changing semantics.

H.3 Calibration Identity

The calibration identity should hash the source checkpoint, candidate adapter class, adapted parameter names, optimizer and learning rate, update steps, evidence schema, benefit-model hyperparameters, feature scaler, residual construction, α, and inference-unit definition. The radius belongs to that identity. Reusing it after a candidate or evidence change is not a benign software optimization; it changes the random variable whose residual is being covered.

H.4 Runtime Accounting

A deployment report should time capture/preprocessing, frozen inference, candidate adaptation, candidate inference, evidence extraction, benefit prediction, interval decision, copy creation, and rollback separately. Mean and tail latency should be reported at the decision-window level. Peak memory should include both model states and optimizer buffers. The compact manuscript leaves these cells unfilled because the canonical cross-dataset manifest does not provide one comparable runtime measurement. Omitting an unsupported number is preferable to quoting controller-only latency as end-to-end cost.

I Evidence Inventory

I.1 Protocol-Specific Feature and Model Schemas

The controlled-grid implementation records 11 label-free coordinates. Two exact schema relations imply row rank at most nine. The previously reported effective rank nine remains an unverified empirical assertion here without a matrix-specific receipt and numerical tolerance; it is not inferred from the proved upper bound. Two coordinates are deterministic differences:

$$\texttt{entropy\_drop} = \texttt{pre\_entropy} - \texttt{post\_entropy},$$
$$\texttt{pbal\_drop} = \texttt{pre\_pbal} - \texttt{post\_pbal}.$$

The panel therefore contains 11 stored values, not 11 independent information sources. Together they describe confidence, class balance, change from the frozen model, and update size.

The exact common coordinate order is

> `pre_entropy`, `pre_conf`, `pre_pbal`, `post_entropy`, `post_conf`, `post_pbal`, `pbal_drop`, `entropy_drop`, `frac_highconf`, `marginal_KL`, and `update_norm`.

The canonical evidence-contract digest for this ordered schema is `76b3951e77c42bbb1359123abd8d937b087c9c42ba0f14e11e61f9aca132f7f7`. The current controlled cross-fit passes these recorded coordinates directly to the GBRT; it applies no separate feature scaler. Nonfinite evidence or outcomes are rejected, and a split that cannot support the exact rank receives an infinite radius and therefore no commitment.

The shared historical builders use natural-log predictive entropy. They normalize class-marginal entropy by $\log K$, stabilize logarithms at 10^{-9}, define the high-confidence fraction with a fixed candidate-confidence threshold of 0.9, and retain marginal KL on its raw scale. The Office-Home builder uses the analogous definitions with 10^{-12} log stabilization. These constants are part of the recorded schema: changing one creates a new evidence variable even when the coordinate name and dimension remain unchanged.

Table 22: Evidence-schema identity crosswalk. Digests bind ordered names and dimensions for the archived contracts; the CCT row instead records its explicit trace schema and sealed gate document. Preprocessing and invalid-input behavior are part of the protocol, not interchangeable defaults.

Authority	Dim.	Schema identity	Model-side preprocessing	Invalid or unavailable state
Common controlled/archive	11	`76b3951e77c42bbb1359123abd8d937b087c9c42ba0f14e11e61f9aca132f7f7`	raw coordinates to candidate-specific GBRT	controlled cross-fit fail-closes malformed or nonfinite matrices
Camelyon17 OOD	17	`bf6eb06e5048705c02293c85038c732e0aacc6e6737f6cb481b3bc5df83ca746`	raw coordinates to archived GBRT	record validator rejects wrong width or nonfinite values
Office-Home	18	`a363c05a2475f3170976b8f7efc30763859e29b9cfe1714f9c4f91a4f1e892bc`	raw coordinates to archived GBRT	absent source logits use recorded zero fallbacks for energy shift and ATC
RxRx1	11	names unrecovered; `8444fd58264350b2379bff08f5ccc0259f381265db7a03f9e0b48415e562805e`	archived model only; coordinate names are not inferred	validator rejects wrong width or nonfinite values
CCT-20	11	`kbound_cct20_label_free_trace_features_v1`; sealed gate document `74db83cca3641c505735bcc9db4f76c588148f1ebb4d7d7c2b713bfbb6eda0e7`	fit-location mean and population-SD standardization before ridge	exact-schema, finiteness, or support failure returns ABSTAIN

Natural-shift artifacts do not all use the same schema. The Camelyon17 OOD record uses the common 11 coordinates plus `disagreement_rate`, `entropy_gap`, `energy_shift`, `bn_kl`, `atc_acc_est`, and `conf_drop`; its 17-coordinate digest is `bf6eb06e5048705c02293c85038c732e0aacc6e6737f6cb481b3bc5df83ca746`. The separate Camelyon17 B–v2 diagnostic uses the common 11-coordinate schema. Office-Home adds `disagreement`, `energy_shift`, `bn_kl`, `atc_acc_est`, `conf_drop`, `prior_kl_source`, and `cand_disagree_mean`; its 18-coordinate digest is `a363c05a2475f3170976b8f7efc30763859e29b9cfe1714f9c4f91a4f1e892bc`. The RxRx1 authority records dimension 11 and digest `8444fd58264350b2379bff08f5ccc0259f381265db7a03f9e0b48415e562805e`, but explicitly marks the coordinate names as unrecovered. We therefore do not silently assign it the common names. The iWildCam record is withheld from release-level numerical claims because its archived scorer used scikit-learn macro-F1 rather than the official WILDS label-present macro-F1 contract. Its canonical evidence metadata nevertheless recovers the common ordered 11-coordinate schema and digest above.

A fitted model and its radius remain specific to their recorded evidence schema. Missing-input behavior is protocol-specific. The current controlled cross-fit fail-closes malformed or nonfinite numeric matrices to ABSTAIN, and CCT-20 returns ABSTAIN on an exact-schema or finiteness failure. The maintained natural-shift record validator instead rejects wrong-width or nonfinite records before routing; historical feature builders encode unavailable BatchNorm statistics as zero, and the Office-Home builder also defines zero fallbacks for absent source logits in its energy-shift and ATC coordinates.

The controlled and archived GBRT protocols use scikit-learn's `GradientBoostingRegressor` (Pedregosa et al., 2011) with squared-error loss, 250 trees, depth 2, learning rate 0.05, subsample 0.8, and random seed 0. The fit is specific to each dataset and adapter. CCT-20 instead uses its sealed ridge model, as described in the main-paper design in Section 7.2. Thus "adapter agnostic" describes the interface: any adapter can propose a candidate that is evaluated and either committed or rolled back. It does not mean that a benefit model fitted for Tent transfers validly to EATA, SAR, or another adapter.

CCT-20 trace and ridge contract. CCT-20 uses a separate ordered trace schema:

> `frozen_mean_entropy`, `tent_mean_entropy`, `entropy_change`, `frozen_mean_confidence`, `tent_mean_confidence`, `confidence_change`, `prediction_disagreement`, `marginal_jensen_shannon_divergence`, `normalized_predicted_class_effective_count`, `normalized_tent_update_norm`, and `batchnorm_batch_source_statistic_divergence`.

The sealed gate identifies this contract as `kbound_cct20_ridge_gate_v1`; the release manifest binds its canonical document by SHA-256 `74db83cca3641c505735bcc9db4f76c588148f1ebb4d7d7c2b713b fbb6eda0e7`. Every feature must be present exactly once and finite. Fit-location means and population standard deviations standardize the coordinates; a zero standard deviation is replaced by one. The ridge penalty is 10, the intercept is unpenalized, and calibration at $\alpha = 0.10$ takes the maximum absolute residual across checkpoints within each location before applying the exact-rank rule over nine locations. A support or schema failure returns ABSTAIN; Mahalanobis distance is diagnostic and never changes the action.

Before ridge standardization, the CCT trace extractor normalizes each entropy by $\log 16$, the Jensen–Shannon divergence by $\log 2$, and predicted-class effective count by 16. It defines the update coordinate as

$$\frac{\|\theta_{\text{after}} - \theta_{\text{before}}\|_2}{\max\{\|\theta_{\text{before}}\|_2, 10^{-12}\}}$$

over the official Tent BatchNorm-affine parameters. The final coordinate is a channel-weighted Gaussian KL divergence from the probe-batch statistics to the stored source BatchNorm statistics, using the layer's BatchNorm epsilon. Probability values are clipped only at machine tiny before a logarithm. These normalizations are distinct from the archived GBRT coordinates above.

Table 23: Non-empirical feature-family summary for the maintained implementation. Each row names one feature family, the state required to compute it, and its protocol-specific validation or fallback. All quantities aggregate unlabeled inputs, frozen/candidate outputs, or the proposed update; the table reports no performance result.

Family	Representative coordinates	Required state	Failure handling
Predictive uncertainty	mean entropy, confidence, quantiles	frozen/candidate softmax	reject NaN or malformed probabilities
Marginal balance	normalized class-marginal entropy and change	predicted-class marginals	fixed class count and normalization
Model change	entropy gap, confidence drop, prediction disagreement	paired frozen/candidate outputs	require aligned sample order
Collapse signal	fraction above adapted confidence 0.9	candidate probabilities	threshold fixed before scoring
Distribution shift	free-energy shift, batch/running-statistic divergence	frozen logits and source statistics	protocol-specific validation or documented zero fallback
Update magnitude	parameter-update ℓ_2 norm	shadow candidate delta	configuration hash must match calibration
Source-calibrated proxy	ATC-style threshold estimate	source-fixed threshold	target labels never used

These protocol-specific dimensions are part of the fitted estimator contract, not universal sufficient statistics. Feature selection may improve prediction, but it cannot defeat the label-kernel construction in the main-paper Lemma 1 over an unrestricted target fibre.

J Implementation Failure Modes and Conservative Responses

The implementation must distinguish a supported negative interval from an unavailable assessment. The deployment contract must preserve the frozen service on an unavailable assessment. Some interfaces return ABSTAIN for invalid features, while malformed authority documents can fail-stop with an integrity exception; the caller must record that failure and must not relabel it as certified FREEZE. Recovery requires a predeclared rule that restores a validated state. The CCT-20 Mahalanobis distance is diagnostic and is not an action trigger; a declared support test is a separate condition.

Table 24: Non-empirical deployment checklist. Each row maps one detectable operational failure to the conservative response required by the implementation contract; it reports no benchmark outcome.

Failure	Detection	Response
Checkpoint/adapter mismatch	stored hash mismatch	rollback and recalibrate
Predeclared support-rule violation	declared support test	withhold commitment
Missing/invalid feature	schema and finite-value check	retain f_0; abstain
Insufficient calibration size	rank $k > n$	infinite radius
Batch too small	minimum-size contract	wait or abstain
Residual transfer concern	group/residual diagnostic	withhold claim; recalibrate or justify transfer
Continual-state drift	state and stream monitor	restore the last validated state

The audit log should store a timestamp, dataset or stream identifier, checkpoint and adapter hashes, evidence schema version, evidence vector, benefit estimate, radius, interval endpoints, action, rollback outcome, latency components, and configuration hash. Labels, if later revealed, are joined in a separate offline audit table. This separation makes it possible to verify that the live decision was target-label-free.

K Prospective Natural-Shift Confirmation Protocol

CCT-20 completed a cell-outcome-unopened, internally sealed natural-shift evaluation but produced all-frozen fail-closed service, not selective routing (Section G.1). So2Sat-LCZ42 then stopped at its development-feasibility gate before gate calibration or target access (Section B.10). Its city-level outcome summaries are now open, and its declared city-level target action conflicts with the runner's checkpoint–city action unit. A new confirmation must therefore be versioned independently, exclude those opened outcomes from tuning, and resolve the action unit before target execution. The design below is future evidence, not a reinterpretation of either completed record. The painting-development evaluations and their queue and BN follow-ups are also already opened and excluded from confirmation.

K.1 Lock Before Target Labels

The protocol should name a fixed, predeclared set of independently trained source checkpoints, one candidate adapter, one evidence schema, one benefit-model class, one calibration rule, one α, and one target-domain family. Adapter hyperparameters must be selected on development domains and then frozen. The lock should hash the checkpoints, code revision, configuration, calibration records, and planned scoring script. It must also name the action and aggregation units, the test unit, and the multiplicity family. A later change creates a new protocol version; it does not overwrite an earlier development-stopped or incomplete arm.

Natural heterogeneity should be built through independently defined environments rather than by pooling whichever benchmark conditions produce opposite signs. Examples include hospitals, camera locations, laboratory batches, acquisition days, physical devices, or visual domains. Environment membership may be observed at deployment, but the measured cell-benefit label remains hidden. The target test environments should be absent from estimator fitting, residual calibration, threshold tuning, and candidate selection.

K.2 Three-Way Data Separation

Development data fit h_θ and select the evidence pipeline. A separate calibration set produces residuals for Eq. (14). The held-out target set is opened exactly once after all software and configuration hashes match the lock. Splitting adjacent frames, patches, or samples from the same session across these stages is leakage even when filenames differ. The split must occur at the environment or session level that defines deployment exchangeability.

The target stream is replayed identically for always-freeze, always-adapt, simple gates, KGA without a radius, and the full empirical benefit-interval gate. Model randomness should either be frozen for paired replay or repeated through independently trained checkpoints declared in advance. Every policy receives the same ordered windows. Labels are stored behind a separate audit interface and are unavailable to the live decision process.

K.3 Endpoints and Decision Rule

The primary endpoint is paired regret against each fixed policy. Success requires both baseline minus KGA regret gaps to be positive and both predeclared intervals to exclude zero, with coverage justified for the declared estimand and sampling model. If the intended claim concerns population risk rather than measured cells, the protocol must also justify that transfer. The false-adapt endpoint is $\mathrm{FA_u}$ with the adapt count reported beside it. Secondary endpoints are $\mathrm{FA_c}$, adapt rate, freeze rate, abstain rate, commitment rate, task accuracy, and latency. Conditional false-adapt is descriptive unless a separate selective-risk result is proved.

The primary benefit-interval rule remains unchanged: adapt if $\widehat{\Delta} - \varepsilon > 0$, freeze if $\widehat{\Delta} + \varepsilon < 0$, and abstain otherwise. The experiment must not lower ε, change α, or select a different benefit regressor after observing target labels. Exploratory sensitivity analyses may be reported, but they remain outside the primary bar.

Table 25: Protocol requirements for a future confirmatory natural-shift KGA experiment. Each row states what must be frozen before target labels are revealed and the corresponding audit record; the table is a design specification, not a completed-study result.

Component	Must be fixed	Audit evidence
Model and candidate	checkpoint, adapter, optimizer, steps, adapted parameters	hashes and immutable configuration
Evidence and estimator	schema, normalization, model class, hyperparameters, seed	feature contract and fitted-artifact hash
Calibration	split unit, residual construction, exact rank, α	calibration IDs, residuals, radius, no target overlap
Target evaluation	environments, windowing, order, exclusion rules, transductive/inductive candidate semantics	sealed manifest and one-time scoring log
Primary inference	two paired regret gaps, cluster unit, interval method	code and random seed fixed before label reveal
Safety	$\mathrm{FA_u}$ and adapt exposure	action log joined to labels only offline
Multiplicity	all candidate–track arms in family	time-stamped inventory including failures
Runtime	end-to-end components and peak memory	raw timing rows on the target hardware

K.4 Interpreting Outcomes

Four outcomes are scientifically useful. An interval-supported reduction in regret relative to each fixed comparator, with nontrivial ADAPT and FREEZE exposure, supports routing utility under the locked natural-shift class. Matching the better fixed policy with few errors supports conservative commitment but not selective routing. High abstention with low error indicates a conservative interval gate and motivates better evidence or more calibration data. A comparator shortfall fails the predeclared routing-utility criterion; excess false adaptation challenges the corresponding safety or calibration criterion. Both outcomes must remain in the result family. None of these outcomes alone establishes the population band: that conclusion still requires the declared class and its structural assumptions.

L Theorem Dependencies, Counterexamples, and Nonclaims

The compact paper uses three related logical statements. First, the disagreement reduction is an identity under the binary setup. Second, the population frontier is a uniform statement over a rich, declared class. Third, the empirical coverage-to-action implication is marginal and conditional on interval coverage. The population statements concern Δ; the reported empirical calibration concerns Δ^{cell}. Connecting these targets requires an additional validity argument. Neither the assumptions nor the quantifiers are interchangeable.

Why the multiclass interface does not extend the binary theorem. Under multiclass 0/1 loss, $\Delta = \mathbb{P}_T(D)(p_a - p_0)$, where $p_a = \mathbb{P}_T(f_a = Y \mid D)$ and $p_0 = \mathbb{P}_T(f_0 = Y \mid D)$. Only the binary setting forces $p_0 = 1 - p_a$. KGA may predict a multiclass scalar score difference, but that interface does not inherit the binary population identity.

Table 26: Logical dependency map for the paper's formal and empirical statements. Each row lists a statement's inputs, conclusion, and excluded inference; it reports no additional empirical result, and a check in one row does not validate premises in another row.

Statement	Inputs	Conclusion	Not concluded
Disagreement reduction	binary labels, 0/1 loss, $\mu_T(D) > 0$	$\Delta = 2\mu_T(D)(M + \gamma)$	calibration transfer or observability of γ
Interior impossibility	label-free fibre, $\beta > 0$, $\lvert M\rvert < \beta$, class richness	evidence-identical laws with opposite nonzero benefits	every empirical abstention is structural
Closed-band result	strict directional semantics and boundary law	no uniformly sound strict action on $\lvert M\rvert \le \beta$	adapt/freeze have different task risk at $\Delta = 0$
Frontier sufficiency	declared $\lvert\gamma\rvert \le \beta$ and fixed M	direction is fixed when $\lvert M\rvert > \beta$	label-free estimation of β
Coverage-to-action proposition	marginal interval coverage for a declared target B	marginal false-adapt/freeze control for B	conditional error control or risk alignment
KGA benchmark	fitted h_θ, declared residual rule, frozen protocol	measured regret and actions on saved units	population-benefit coverage or universal transfer to a new domain
Lean build	encoded definitions and assumptions	kernel-checked formal statements	correctness of unencoded data and deployment assumptions

Why the boundary is separate. For $|M| < \beta$, the construction chooses two latent residuals that yield opposite nonzero benefits. At $|M| = \beta > 0$, one admissible residual can make the benefit zero while another retains the sign of M; opposite nonzero signs need not exist. Both cases block a uniform strict commitment, but for different reasons. At $M = \beta = 0$, every law in the zero-residual class has zero benefit under the model. Abstention remains the maximal action only under the paper's strict-direction convention, not because either task-risk choice is worse.

Why a calibration residual need not be drift. Let $P_T = P_S$, with equiprobable inputs a, b, deterministic labels $Y(a) = 1$, $Y(b) = 0$, and candidate $f_a \equiv 1$. Set $f_0(a) = 0$, $f_0(b) = 1$, and $s \equiv \frac{1}{2}$. The candidate is correct on half the source inputs, so this constant score is source-calibrated in the ordinary sense. Yet disagreement occurs only at a, where the candidate is always correct. Thus $M = 0$, $\gamma = \frac{1}{2}$, and $\Delta = \frac{1}{2}$, although the source and target laws are identical. Conditioning on disagreement, not a distribution change, explains the nonzero residual. The frontier requires a bound on this residual regardless of its cause. Top-label calibration conditions on the predicted class (Gupta and Ramdas, 2022); neither it nor ordinary score calibration automatically implies calibration conditional on disagreement.

Why coverage is the premise. The coverage-to-action proof is elementary once Eq. (9) holds: on the coverage event, an interval wholly above zero cannot contain a nonpositive Δ^{cell}. The scientific burden is therefore to justify coverage for the declared outcome and calibration/deployment pair. Joint exchangeability of the residual scores is one route; known shift correction or another valid construction may be another. Risk alignment can explain why a useful predictor transfers, but it is not an additional logical premise after coverage is given.

Why cell coverage is not population coverage. Consider the fixed binary predictors $f_0 \equiv 0$ and $f_a \equiv 1$. Let the target input be a with probability 0.45 and b with probability 0.55, with deterministic labels $Y(a) = 1$ and $Y(b) = 0$. Population benefit is $\Delta = -0.10$. In a one-item evaluation cell, a fixed predictor that maps a to $+1$ and b to -1 predicts Δ^{cell} perfectly without reading the new label. With radius zero, cell coverage is perfect, yet the gate selects ADAPT with probability 0.45 despite negative benefit under the fixed population law. This constructed example does not contradict the main-paper Proposition 2; it shows why its covered target cannot be changed.

Why exchangeability alone does not validate sign flips. As a separate null example, let all nine cluster gaps equal the same fair random sign S. This vector is exchangeable, each gap is symmetric about zero, and a global sign change preserves its law. Nevertheless, when $S = +1$, independent-coordinate sign-flip enumeration for the mean gives a reference value of $1/512$. That occurs with probability $1/2$, not $1/512$. The vector fails Eq. (16). This example explains the required assumption; it is not a model fitted to the CCT-20 observations.

Why a null is not an impossibility result. A null benchmark row can arise because the true effects are small, the candidate is one-sided, the evidence map is weak, the benefit model underfits, the radius is wide, or calibration does not transfer. The population impossibility requires evidence-identical target laws inside the declared class with incompatible strict directions. The canonical benchmark table does not estimate that full fibre. It therefore reports weak or adverse empirical regimes without calling them proven unknowable worlds.

What would falsify the practical claim. The practical claim is conditional and testable. A clean target replay with the locked pipeline can record a high FA_{u}, low inclusion, or regret above both fixed policies. A checkpoint or adapter change, or a family-held-out split, can provide evidence of limited transfer or optimistic cell-level cross-fitting. Such findings can reject a predeclared empirical endpoint or provide evidence against transfer or coverage assumptions; they do not by themselves logically refute a marginal guarantee. Poor utility can also coexist with valid coverage. A claim of coverage or transfer failure requires its own appropriately designed statistical assessment; none of these finite outcomes refutes the population identity.

M Formalization and Artifact Scope

The Lean project under `docs/research/kbound/formal/` provides a kernel-checked inventory for the theorem names it exports. The compact paper relies on the LaTeX statements above; it does not claim that every sentence has a one-to-one Lean theorem.

In particular, deployment-class richness, risk alignment of a learned evidence map, exchangeability of a benchmark split, and correctness of data preprocessing remain outside the proof assistant unless separately encoded.

M.1 Proof traceability and limits

The formal evidence is a source-bound, scoped verification record. The 8 September 2026 integration receipt records a successful Lean 4.29.1 strict-core build and transitive-axiom audit for 315 registered declarations: 65 retained core declarations and 250 additional results, including supporting lemmas. The theorem map contains 326 distinct `#check` entries across all namespaces; eleven are outside that registry. Thirteen public-root example files contain 80 examples. These are different inventories, not a percentage of the paper proved.

The source-bound summary `formal/full_report_bridges_verification_20260908.portable.json` links the combined audit, example checks and independent reviews by hash. The build is incremental with cached dependencies, not a cold dependency rebuild. The separate claim census is `formal/full_report_bridges_semantic_20260908.portable.json`. Earlier dated receipts retain their historical scopes. The audited axiom dependencies are `propext`, `Classical.choice` and `Quot.sound`.

Population reduction and frontier. `JointKernelScore.joint_score_benefit` derives the score-residual identity from the conditional kernel of an actual binary joint law. `ActualWorldFrontier.actual_identified_interval` identifies the clipped benefit interval over the full fixed-input-marginal residual class. `ActualWorldEvidence.actual_augmented_fibre_strict_direction_iff` connects this class to a fixed marginal-based observation experiment. The restricted-class results in `ActualWorldSubclass` require the legal interior opposite-pair and boundary zero constructions; individual directional uniqueness additionally requires a nonempty class. These results do not establish those membership conditions for an arbitrary deployment subclass or justify conditioning on a model pair adapted on the same observed batch.

Audit floor. `measurable_audit_floor` proves the bound at the possibly unattained supremum of a nonempty bounded residual fibre under a common observation law. `ActualFibreRadius.actual_augmented_fibre_radius` supplies the actual full-class radius equality under positive disagreement mass, $0 \leq \beta \leq \frac{1}{2}$, and the fixed observation experiment. This is not an estimator of a valid deployment β from unlabeled data.

Coverage and decision. `ExtendedRadiusCertificate.directional_error_probability` proves that coverage for the named scalar target bounds the union of false strict directions, allowing measurable finite or infinite radii. `ExactConformal.literal_residual_certificate` constructs the literal ceiling-rank absolute-residual interval, including ties and the infinite-radius case, under joint score exchangeability. Neither result establishes benchmark exchangeability, calibration transfer, or operational label custody. For a population-target application, valid coverage of population benefit is the required statistical premise; exact fibre-wide identification is a separate question. The `PopulationTransfer` module combines cell and sampling coverage by a union bound, but does not supply the reported experiments with a population sampling-error radius.

Supporting statements. `WeightedHelpful` assumes per-cell helpfulness and common nonnegative weights, with positive total weight for means. `FeatureRank` proves a nine-dimensional real-linear schema and row rank at most nine under the exact coordinate relations; it does not establish observed numerical rank or probabilistic independence. `RiskAlignment` proves that alignment is weaker than strict identification, including an actual zero/positive-law witness.

Full-report implications. `InformationRefinement` covers null-disagreement risk equality and refinement of actual joint-observation fibres, with nonempty/bounded premises for real radii.

`ConditionalExposure` derives conditional error bounds from marginal error and positive action exposure. `IntegrableProxy` proves the expectation decomposition and sufficient signs with both terms integrable under the same law; it does not prove general-loss exactness.

The current prose states the fixed observation, model-pair, seed-coupling and nonempty-subclass qualifications. Their empirical applicability remains unproved. Manuscript revisions require their own final source binding and semantic review; unchanged Lean sources do not make every revised sentence kernel-checked. The historical full one-bit/H/ratio-rate extension remains excluded. Formal verification, empirical provenance, implementation tests and publication rendering therefore retain separate scopes.

One historical extension remains outside the verified scope. A measurable label swap preserves label-free evidence and reverses population benefit, but selecting one representative from each swap orbit need not identify the sign: distinct orbits can share the same evidence law and have opposite selected signs. Lean verifies this counterexample and the corresponding set-theoretic fibre-consistency criterion, not a general measurable decoder or the older H/ratio-rate extension. Those stronger claims are excluded from this compact submission. The `-full-foundations` gate therefore remains failing for the sixth layer; the scoped kernel audit checks the registered results and rejects nonstandard proof axioms.

The formal build and empirical build answer different questions. Lean checks that a proposition follows from formal assumptions. The reconciliation scripts check that displayed aggregates match saved records under implemented definitions. Neither checks the truth of an unencoded deployment assumption, and neither substitutes for independent experimental replication.

The authoritative compact artifacts are therefore layered:

1. this source and its compiled PDF for the scoped narrative;
2. shared theorem inputs for the exact statements and proofs;
3. canonical JSON and generated LaTeX for numerical results;
4. the source manifest for record-level provenance;
5. package tests and Lean logs for implementation and formal checks.

Historical drafts, dashboards, and exploratory search folders can remain useful development records, but they are not evidence sources for this compact submission unless referenced by the canonical manifest.

N Reproducibility and Release Procedures

N.1 Canonical Numerical Source of Truth

The CIFAR rows and appendix reconciled point estimates share one numerical source: `experiments/kbound/results/reconciled_panels_v1/canonical_panel_results.json`, whose SHA-256 is `35d4c165843de1ece3cb35ffb4ac50dbfcbfa33b646754e2b6d133fbdfa78c6e`. Its input inventory is bound separately by source-manifest SHA-256 `03b1d2b1e9e5ed1cf835126f871ed83eb8497a1ac81727b3955af0d69eb89742`; the canonical JSON also records generator, implementation-dependency, and per-array bindings. Auxiliary artifacts retain narrower roles. The family analysis is the retrospective sensitivity in Table 16; the POEM/AETTA artifact is historical; and the So2Sat record is a locked development stop. The generated result manifest indexes each reported claim and scope. None of these auxiliary records overwrites the canonical point-estimate panel.

N.2 Reproduction Pipeline

The full gate uses Python 3.12 and `requirements-research.txt`. The smaller paper-only profile does not include the Torch imports used during repository-wide test collection. The model runtime uses

PyTorch (Paszke et al., 2019). From the repository root, the release command is:

```
KBOUND_PYTHON=.venv/bin/python bash
docs/research/kbound/runbooks/release_candidate.sh all
```

It runs the read-only preflight, validates and regenerates result authorities, executes software and formal checks, builds the four current role PDFs, renders every PDF page, and emits release checksums. A DOCX may be produced as a nonrelease convenience artifact, but it is not a fifth current role. The gate consumes frozen model outputs and never launches training.

For presentation-only rebuilding from already validated authorities, the command is:

```
SOURCE_SNAPSHOT_COMMIT=<12-hex-source-commit>
BUILD_LONG_TMLR=1 BUILD_SHORT_MAIN=1
BUILD_SHORT_SUPPLEMENT=1 BUILD_FULL_REPORT=1 bash
docs/research/kbound/scripts/build_pdfs.sh
```

This shorter command validates the frozen authorities, regenerates presentation tables and figures, and builds the four maintained role PDFs. It does not replace the full release gate or create new empirical evidence.

N.3 Calibration Identity and Implementation Contract

The maintained API binds an interval-gate record to the checkpoint, adapter configuration, evidence schema, benefit estimator, residual construction, safety level, and scoring unit. The deployment path must fail closed when any identity differs. `KGA.certify_evidence` requires a frozen estimator, the exact schema, a protocol hash, and a separate residual set. The lower-level `certify` function is an audit interface; it does not construct evidence or validate transfer by itself.

N.4 Formalization and Lean Scope

Lean artifacts and validators check the encoded theorem cores and selected finite constructions. Formalization of learning-theory results has precedent, including a Rademacher-complexity generalization bound (Sonoda et al., 2025); we cite its pinned 2025 version as context, not as a proof of K-Bound's assumptions. These checks do not certify dataset correctness, preprocessing, empirical transfer, exchangeability, or every deployment assumption stated in prose. Machine checking strengthens the statements it encodes; it does not validate the benchmark pipeline outside those statements.

N.5 Release Checklist and Audit Logs

The Phase-1 provenance audit matches all 106 canonical source/compact hash pairs, all 10 recoverable configuration hashes, and all 6 archived checkpoint hashes; it also records a 21-file current-code snapshot. Historical runs that did not serialize full identity remain explicitly partial rather than being backfilled. The release gate runs reconciliation tests, claim-ledger and result-manifest validation, manuscript checks, LaTeX builds, checksum verification, and visual PDF inspection. Build logs, the canonical manifest, the provenance seal, and the final PDFs are archived together.

Table 27: Release-stage audit contract. Each stage has an explicit authority and a fail-stop condition, preventing a mixed-generation or partially verified manuscript set from being published as current.

Stage	Checked authority	Failure response
Numerical reconciliation	canonical panel, source manifest, CCT manifest and receipt	stop on hash or claim mismatch
Source binding	reviewed source snapshot and release-role map	stop on missing or ambiguous identity
Compilation	four maintained TeX drivers from one source closure	stop on compile, reference, or overflow defect
Structure and anonymity	page structure, fonts, metadata, links, and anonymous TMLR author fields	reject malformed output or identity leakage
Visual inspection	fresh rendering of every maintained PDF page	repair clipping, blank regions, orphaned captions, or displaced floats
Publication	exact four-PDF inventory and SHA-256 manifest	publish atomically; never mix generations